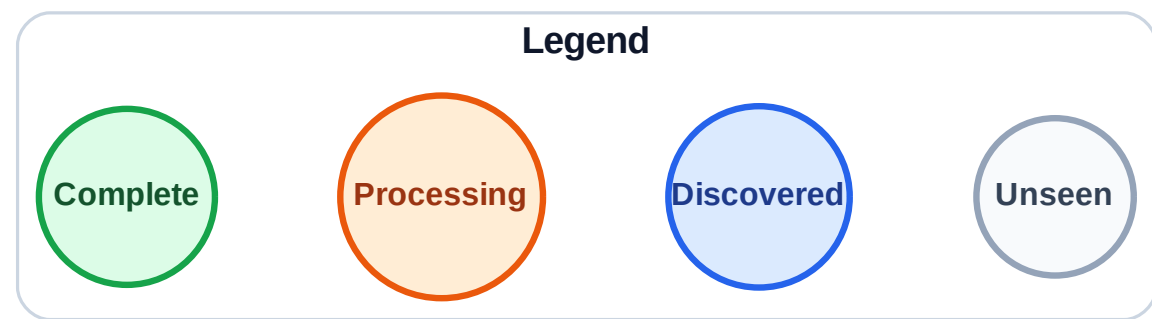


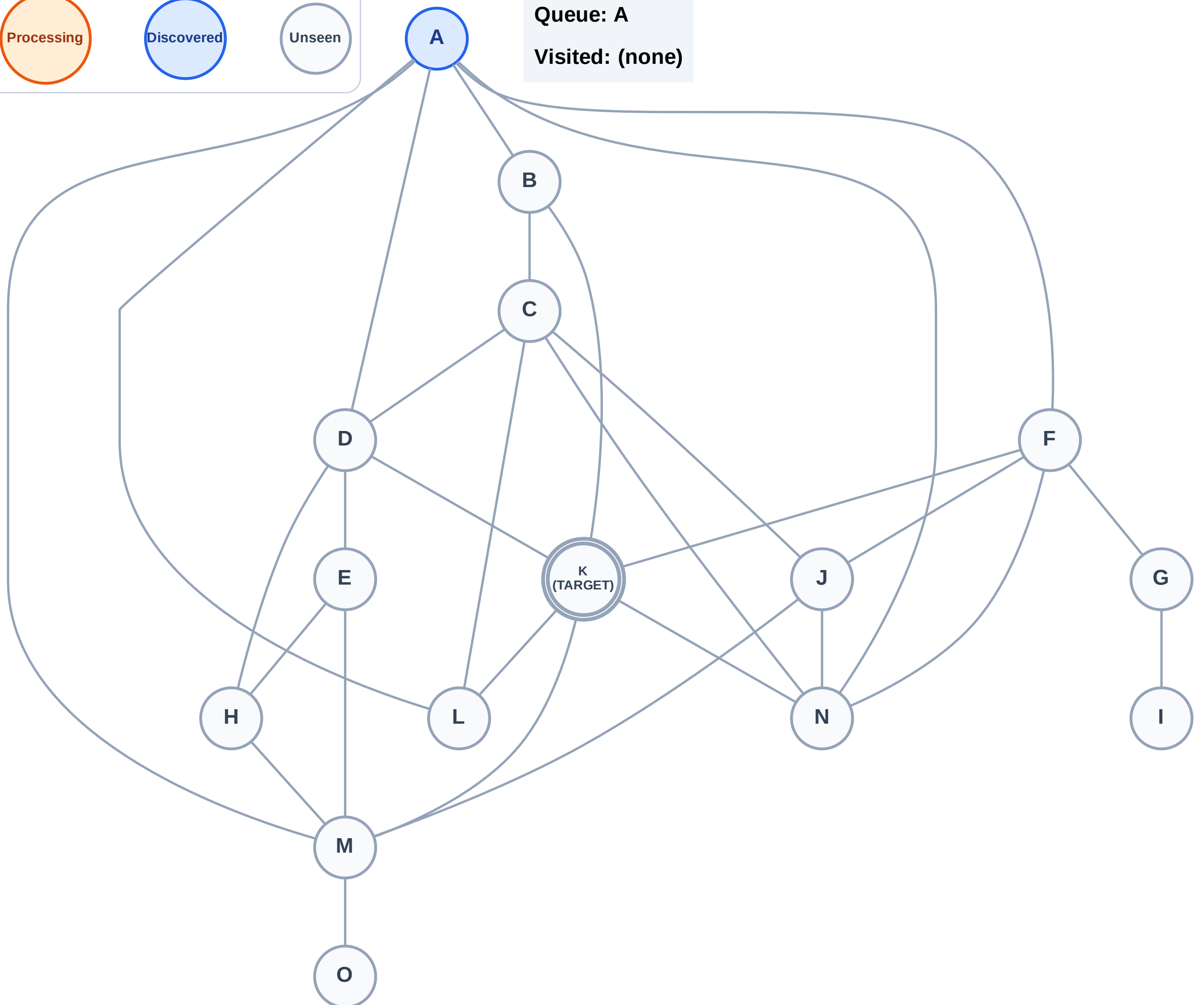
# BFS Traversal

Step 1: Start at A (Seeking target node K)



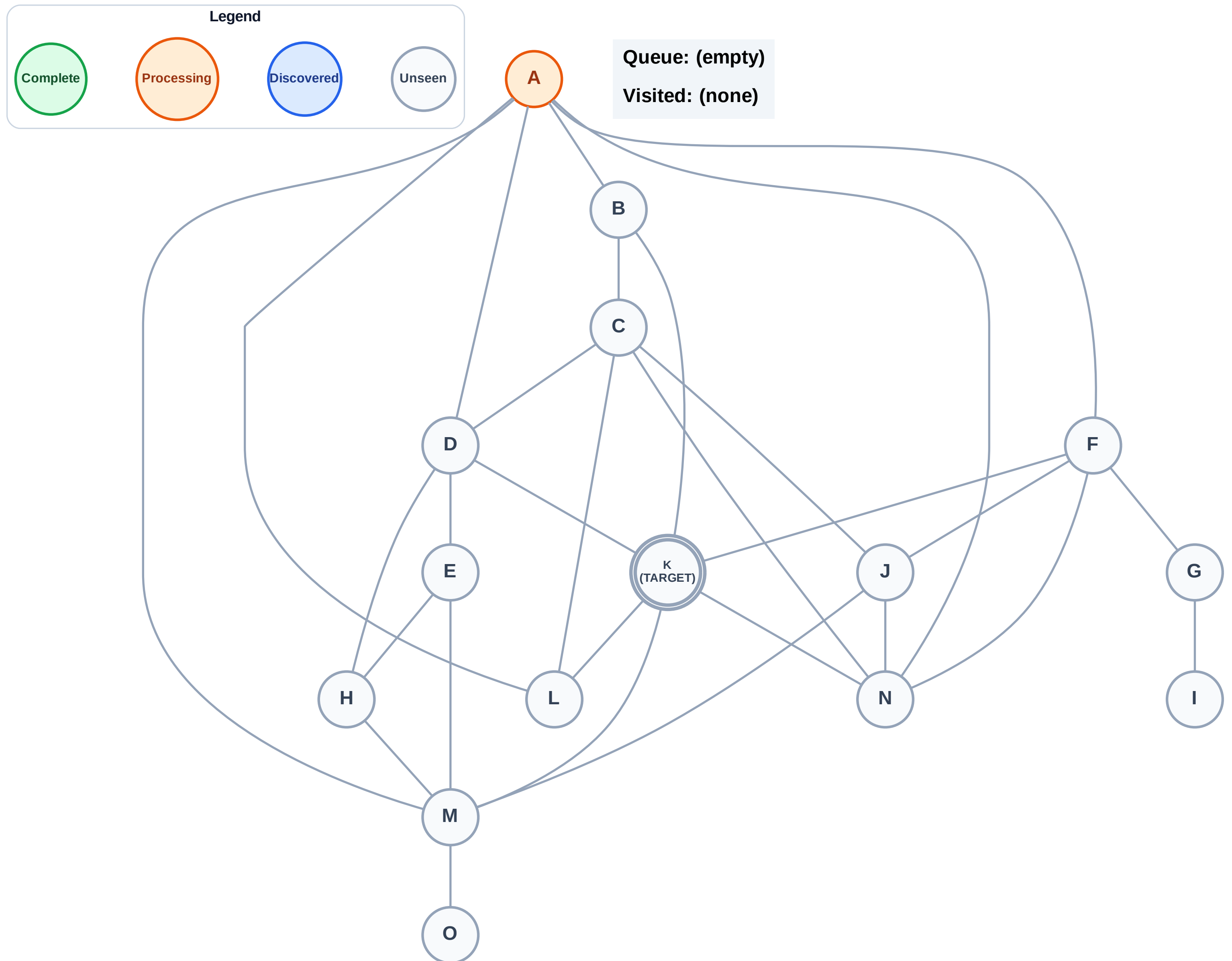
Queue: A

Visited: (none)



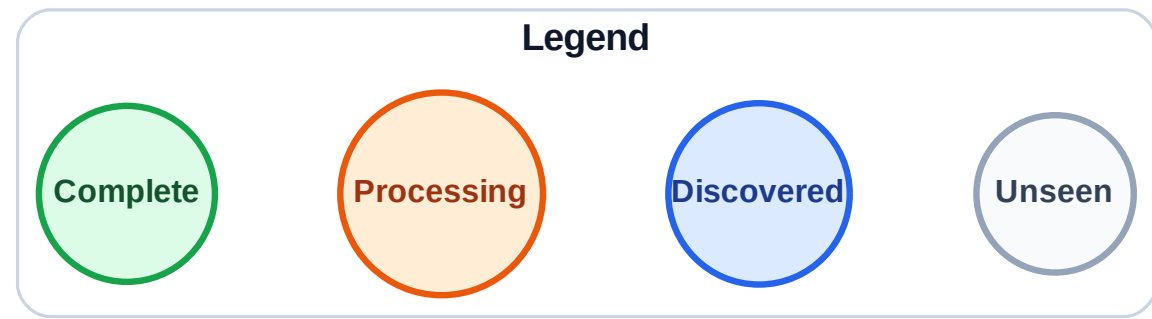
### Step 2: Process node A

### Step 2: Process node A



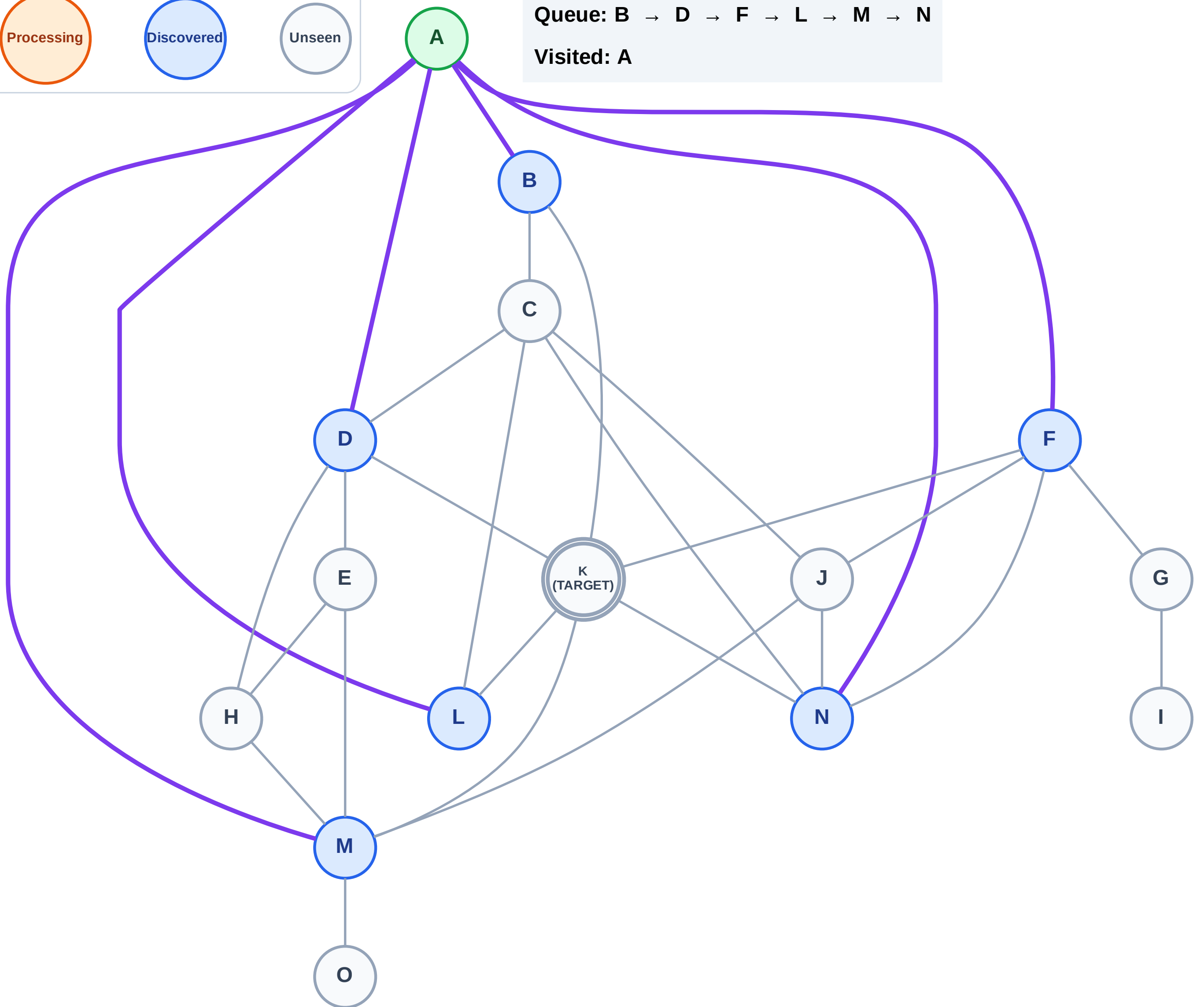
# BFS Traversal

Step 3: Discover B, D, F, L, M, N from A



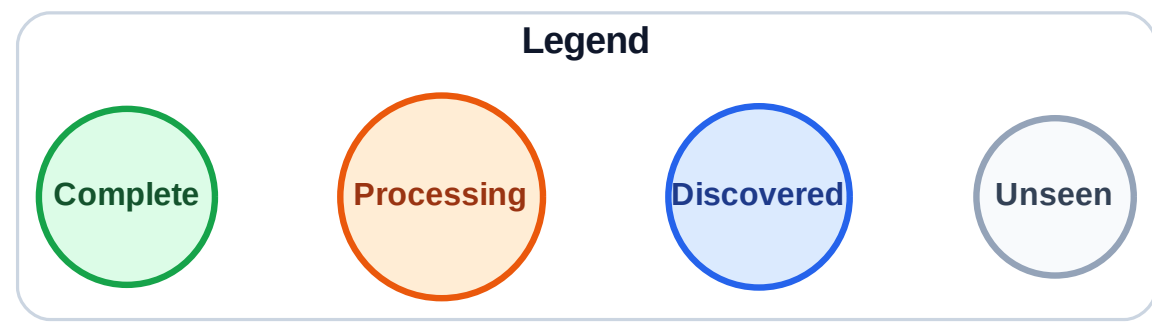
Queue: B → D → F → L → M → N

Visited: A



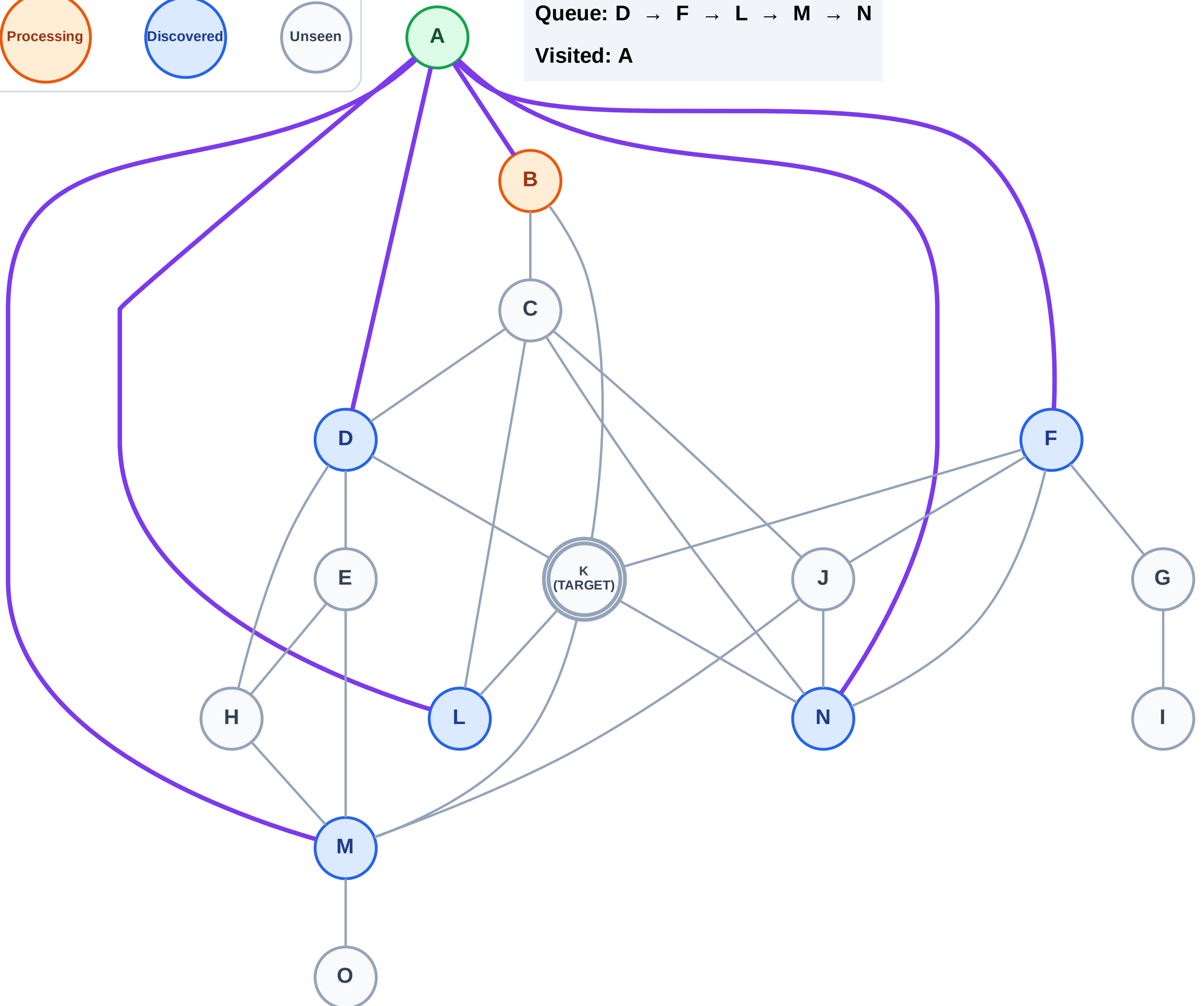
# BFS Traversal

Step 4: Process node B



Queue: D → F → L → M → N

Visited: A

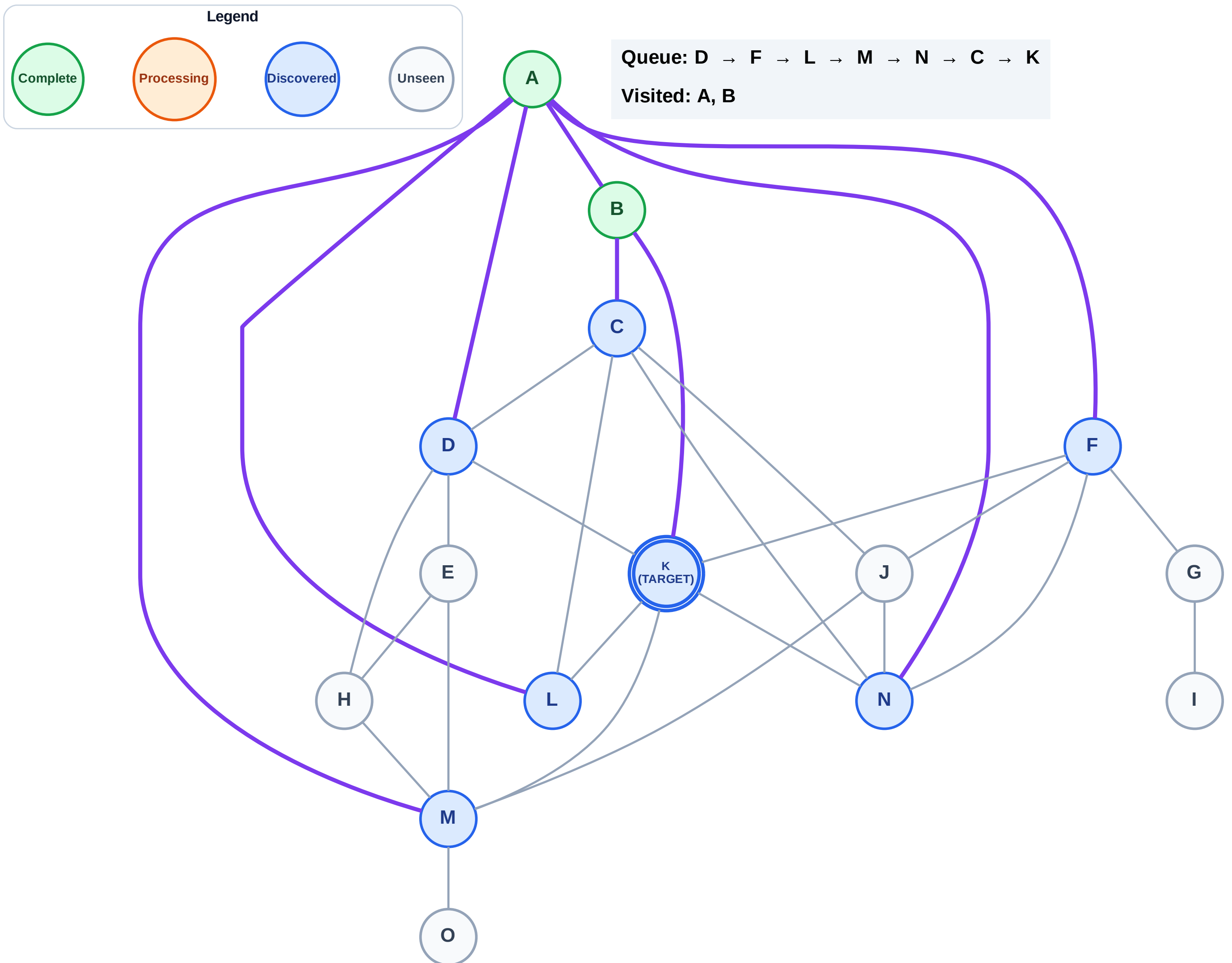


### Step 5: Discover C, K from B

### Step 5: Discover C, K from B

**Queue: D → F → L → M → N → C → K**

**Visited: A, B**



# BFS Traversal

Step 6: Process node D

Legend

Complete

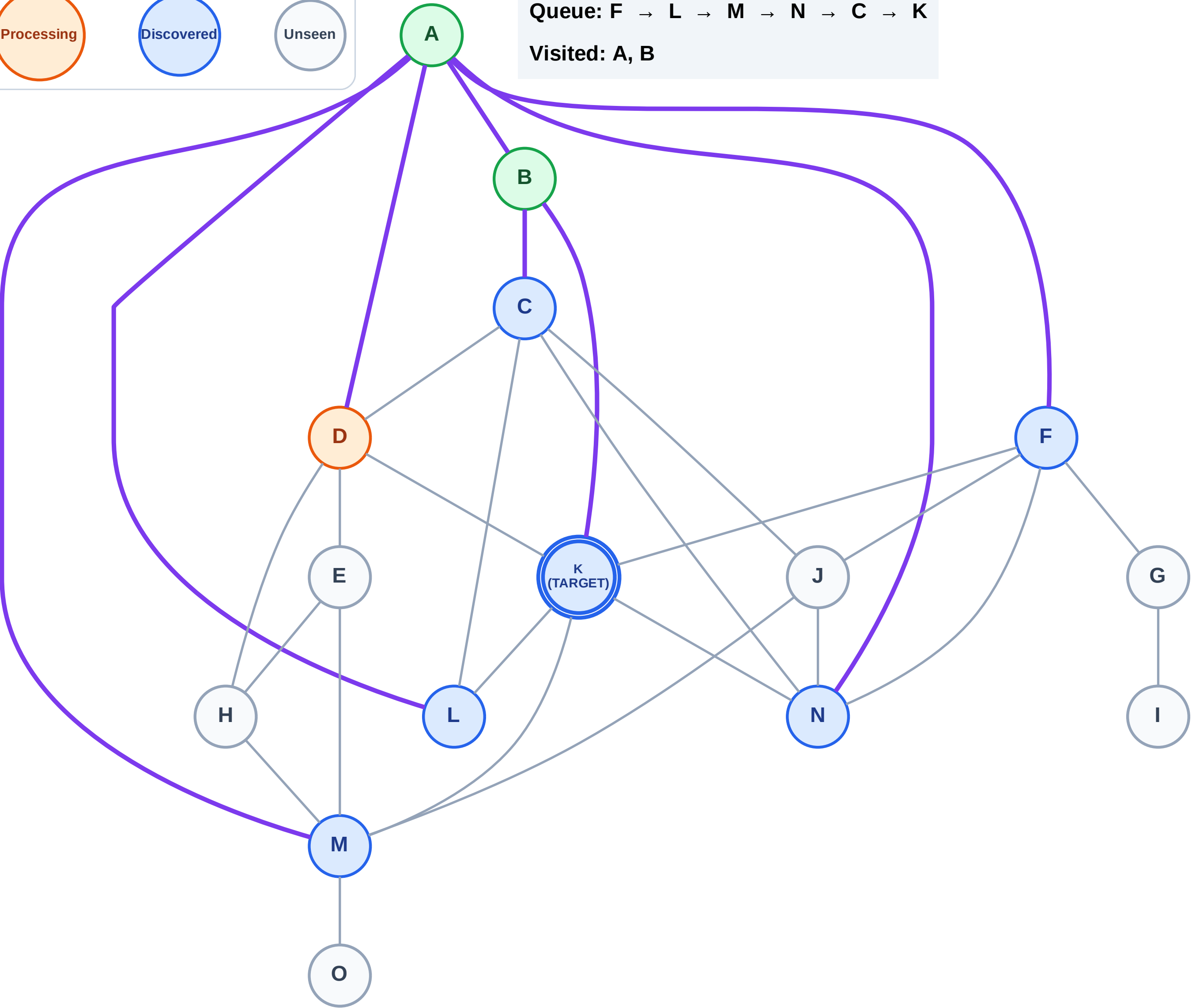
Processing

Discovered

Unseen

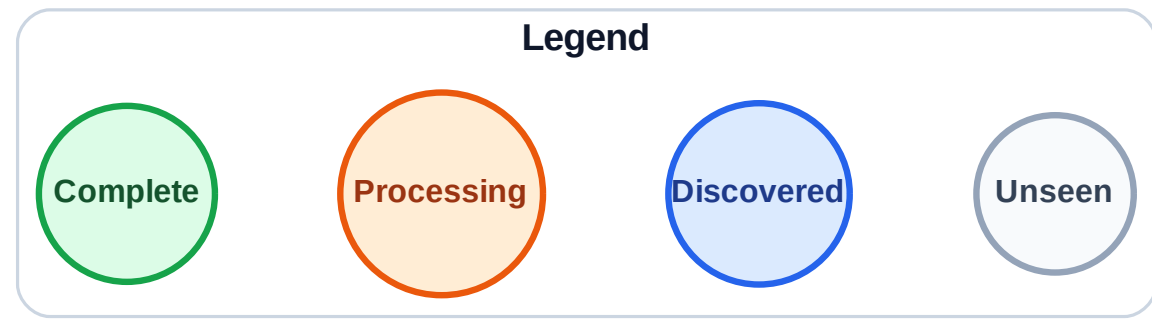
Queue: F → L → M → N → C → K

Visited: A, B



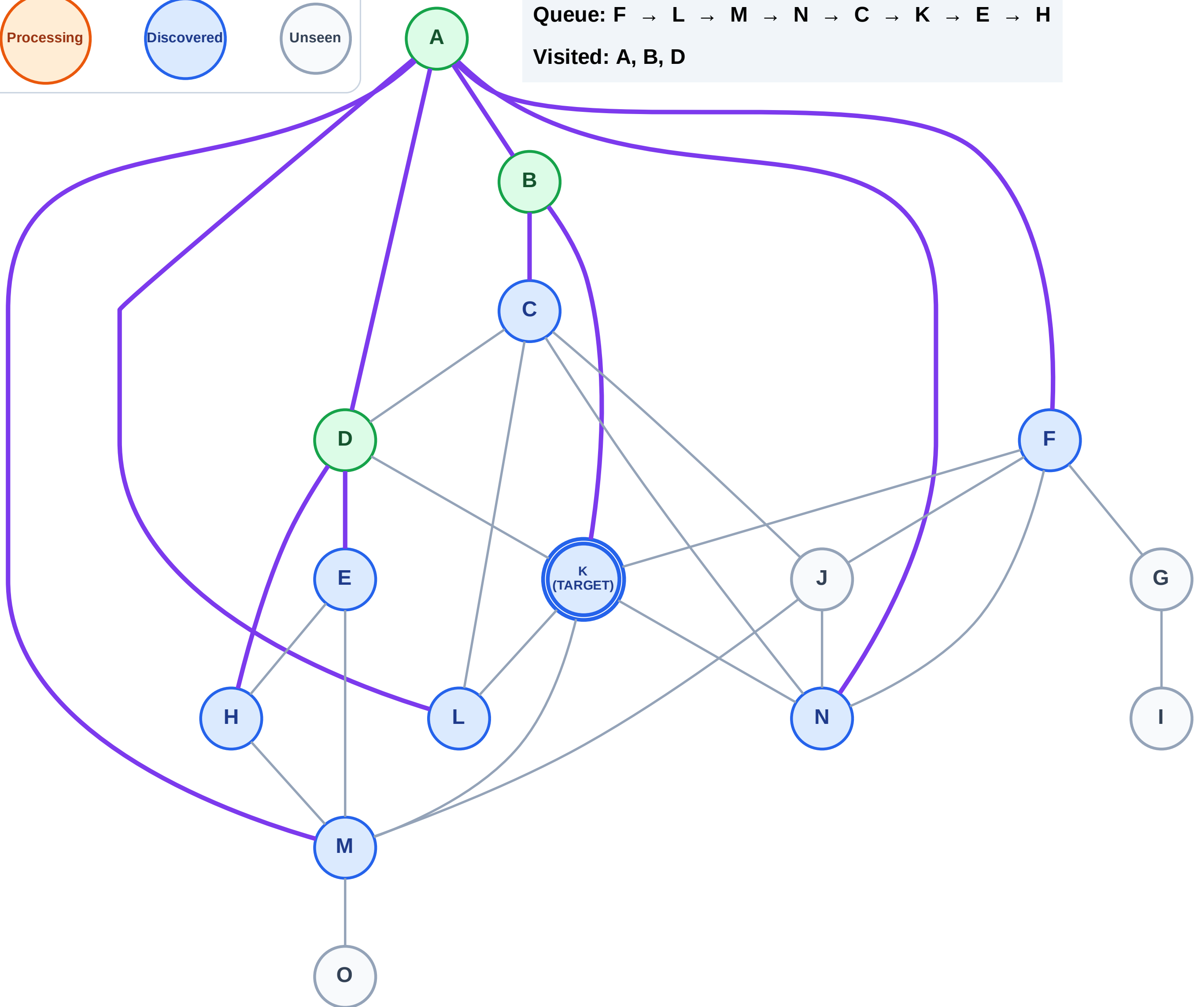
# BFS Traversal

Step 7: Discover E, H from D



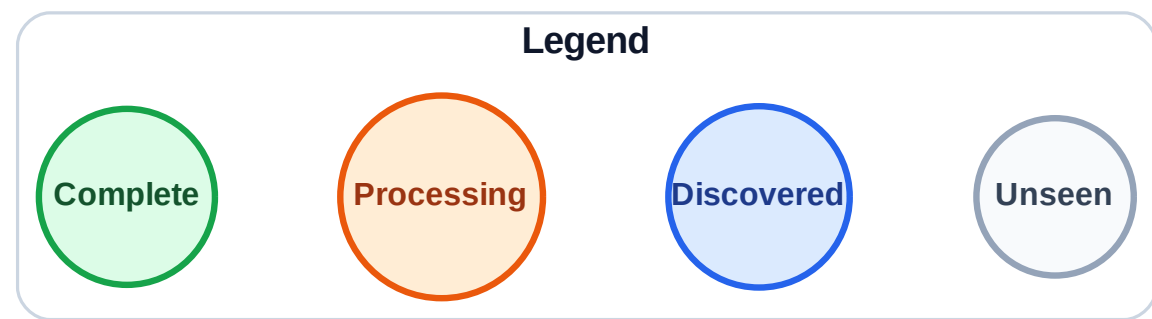
Queue: F → L → M → N → C → K → E → H

Visited: A, B, D



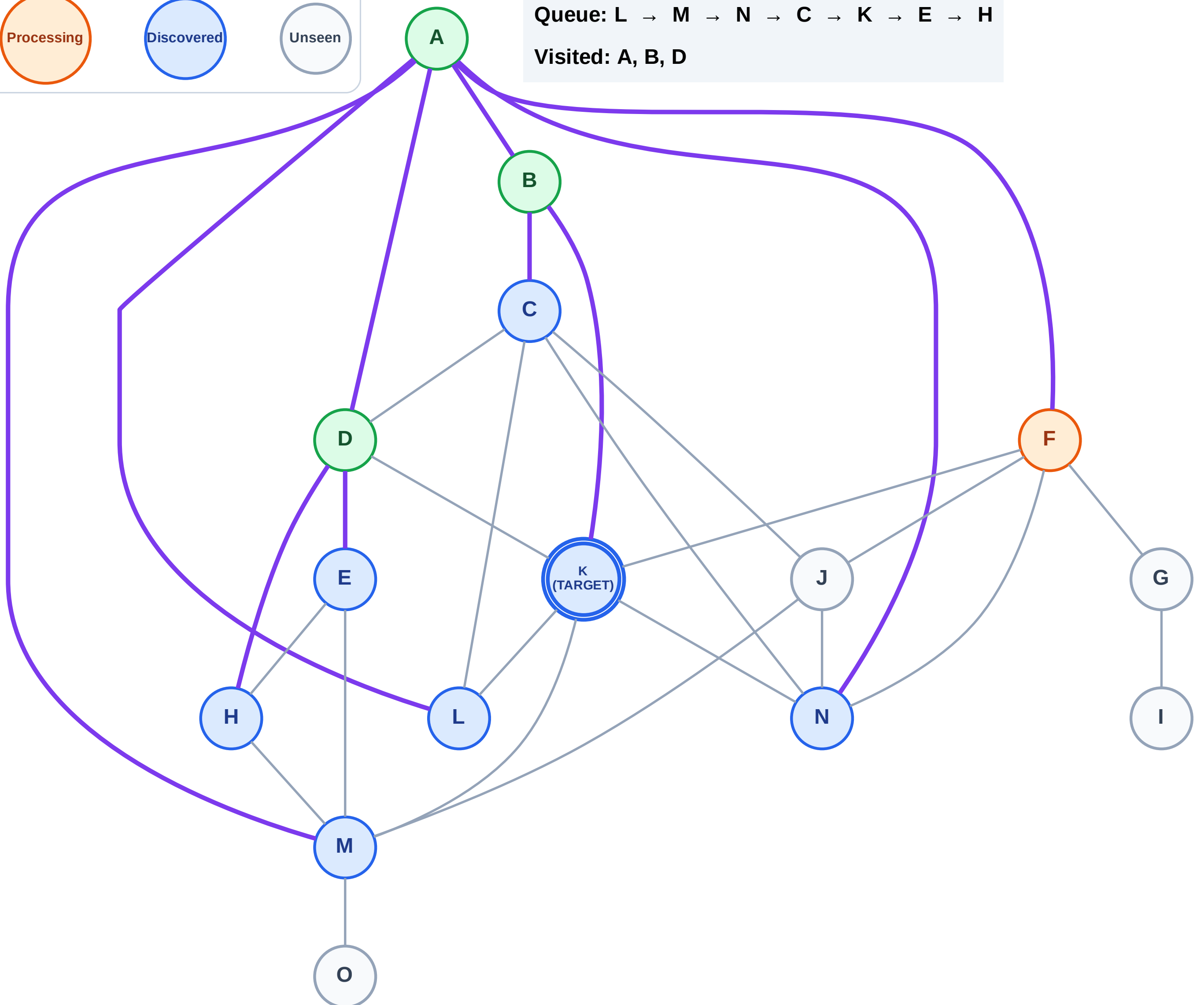
# BFS Traversal

Step 8: Process node F



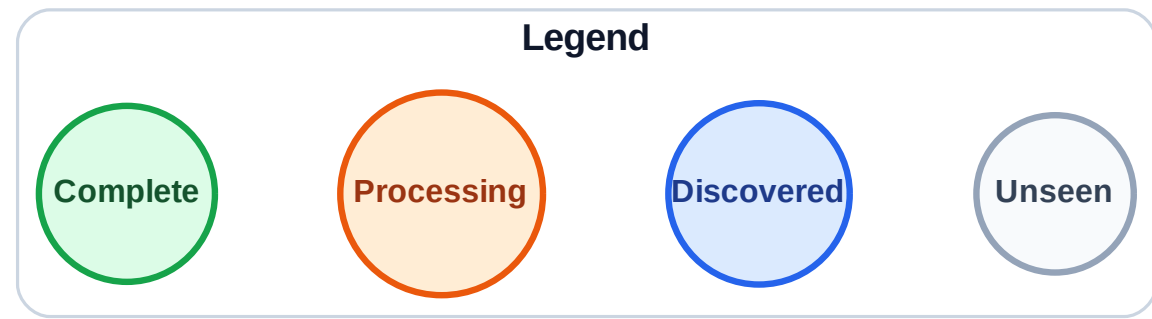
Queue: L → M → N → C → K → E → H

Visited: A, B, D



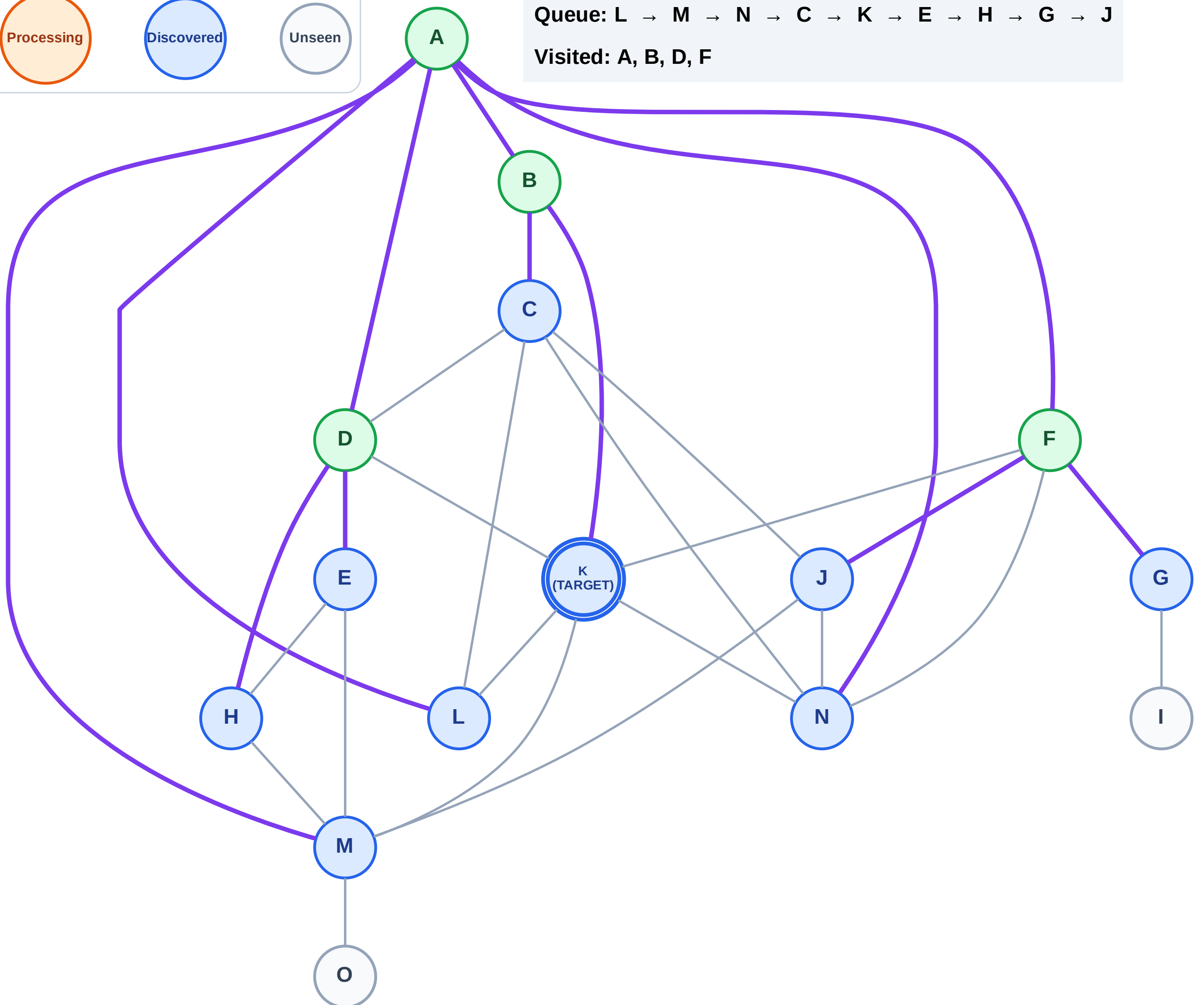
# BFS Traversal

Step 9: Discover G, J from F



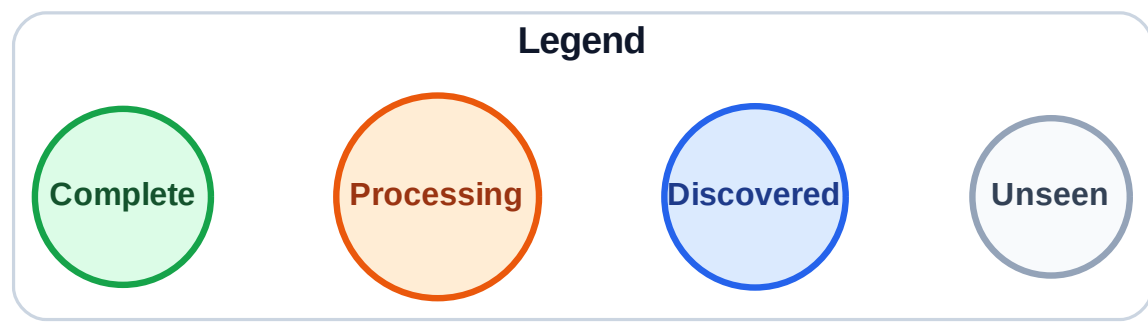
Queue: L → M → N → C → K → E → H → G → J

Visited: A, B, D, F



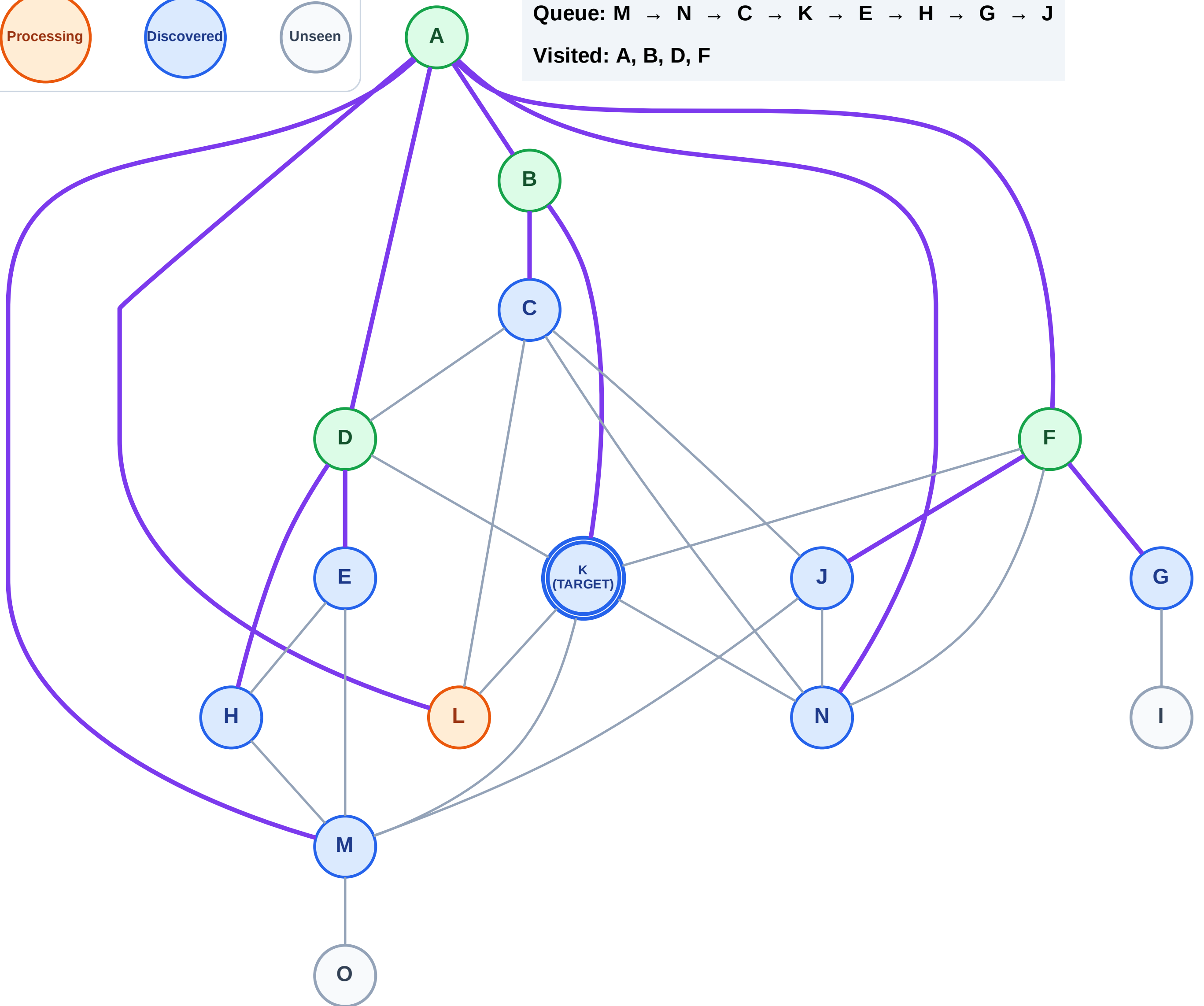
# BFS Traversal

Step 10: Process node L



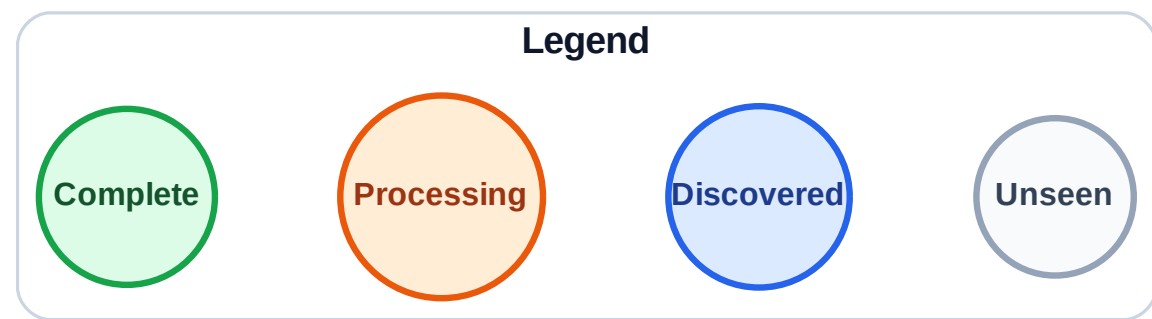
Queue: M → N → C → K → E → H → G → J

Visited: A, B, D, F



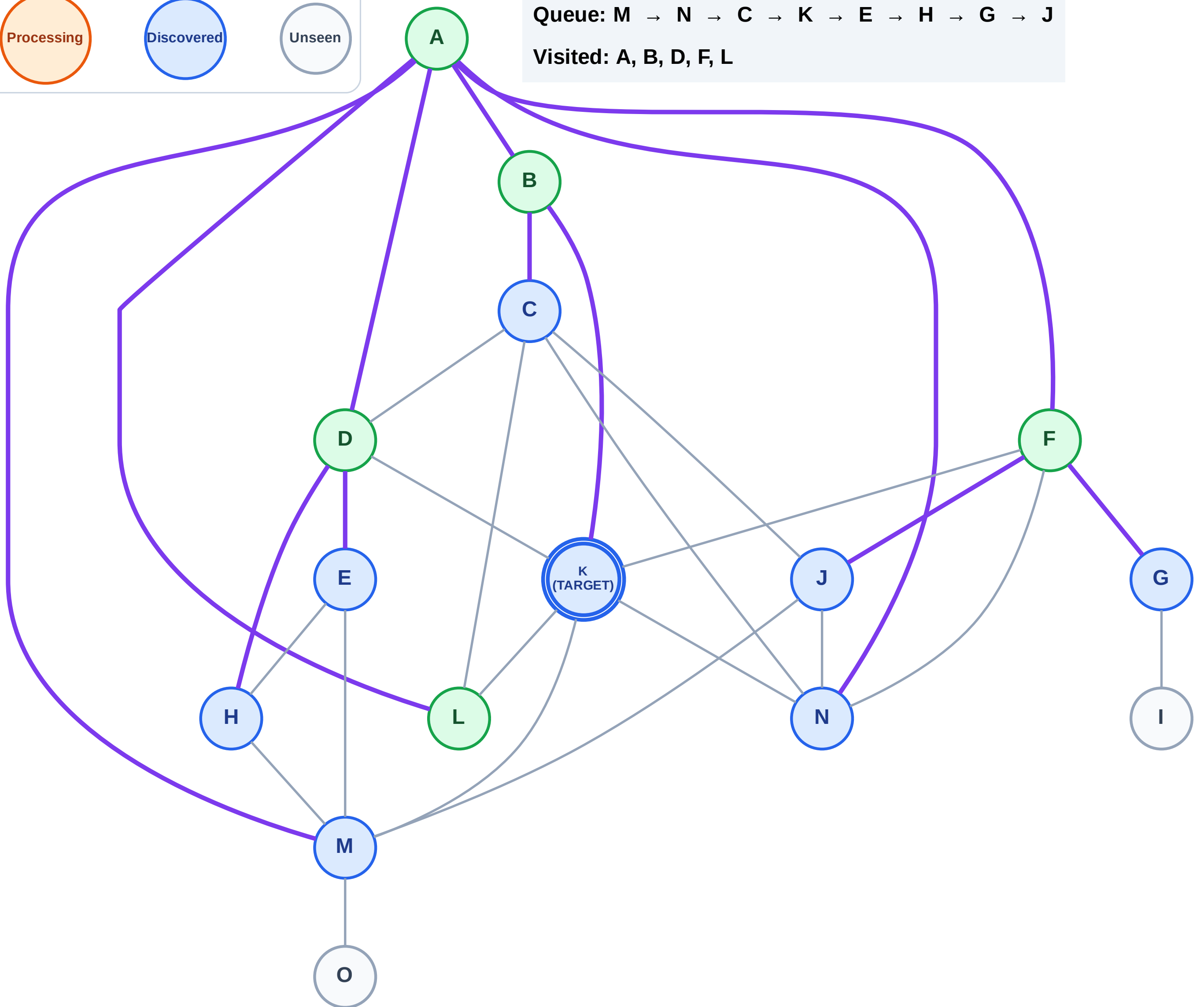
# BFS Traversal

Step 11: No new neighbors from L



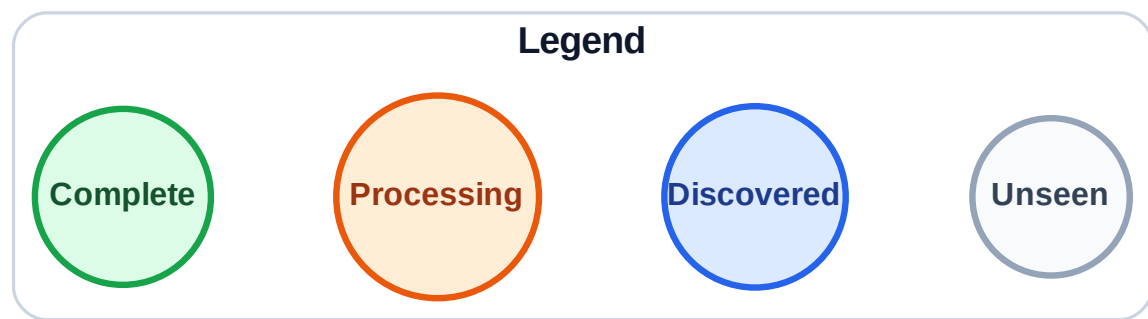
Queue: M → N → C → K → E → H → G → J

Visited: A, B, D, F, L



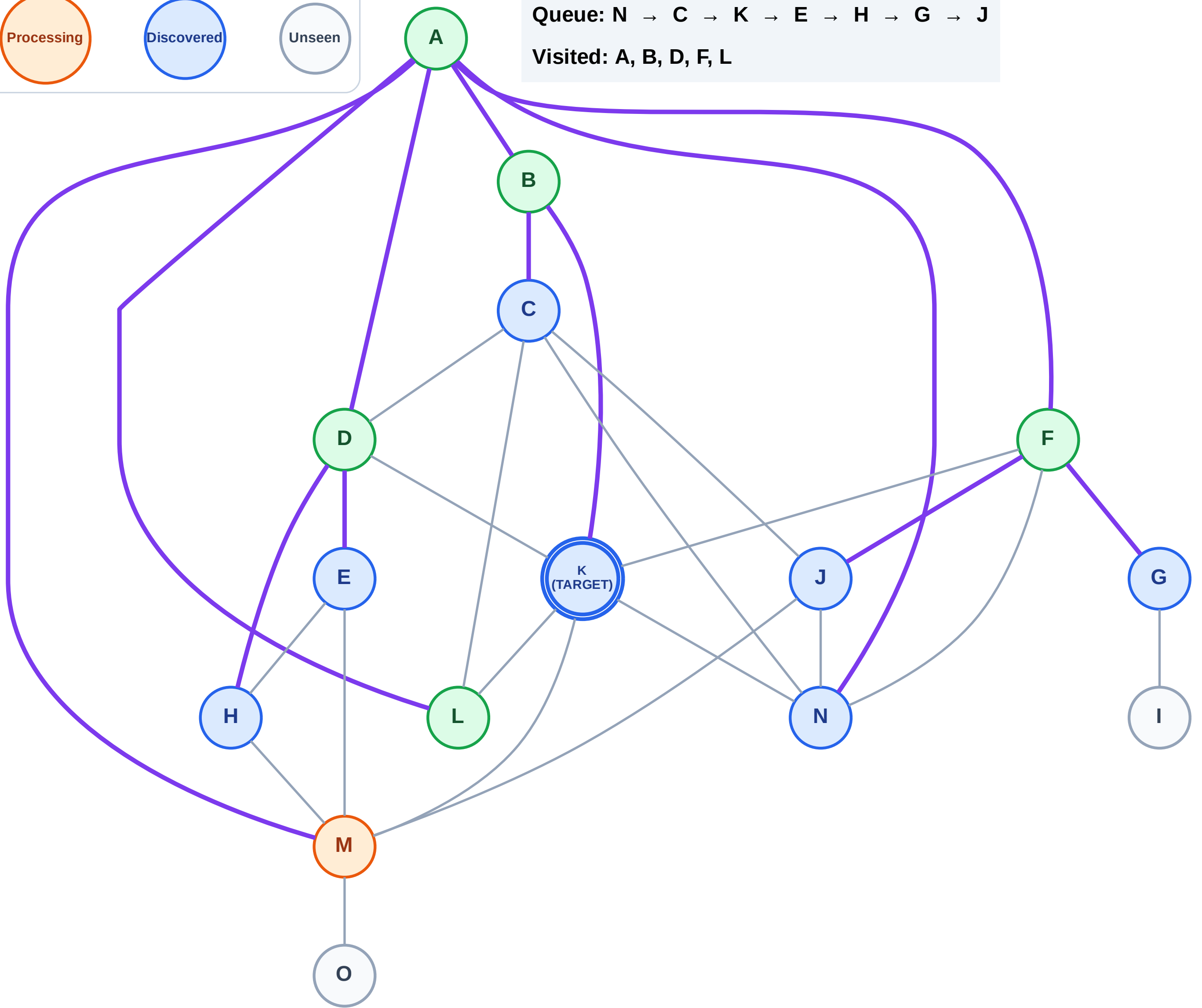
# BFS Traversal

Step 12: Process node M



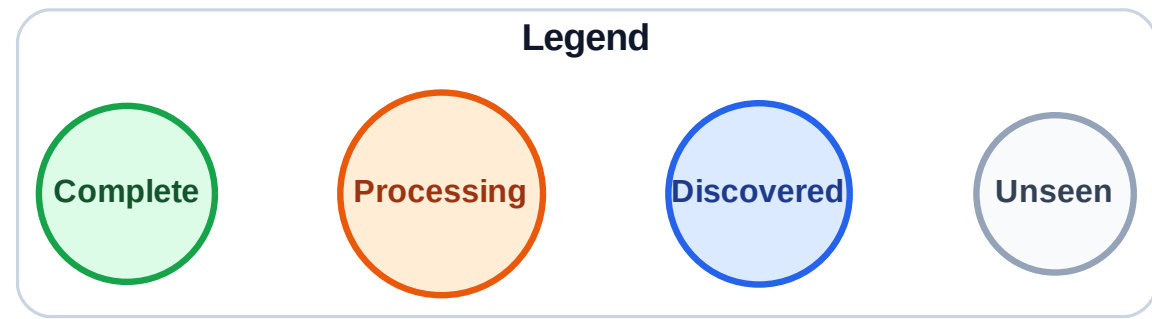
Queue: N → C → K → E → H → G → J

Visited: A, B, D, F, L



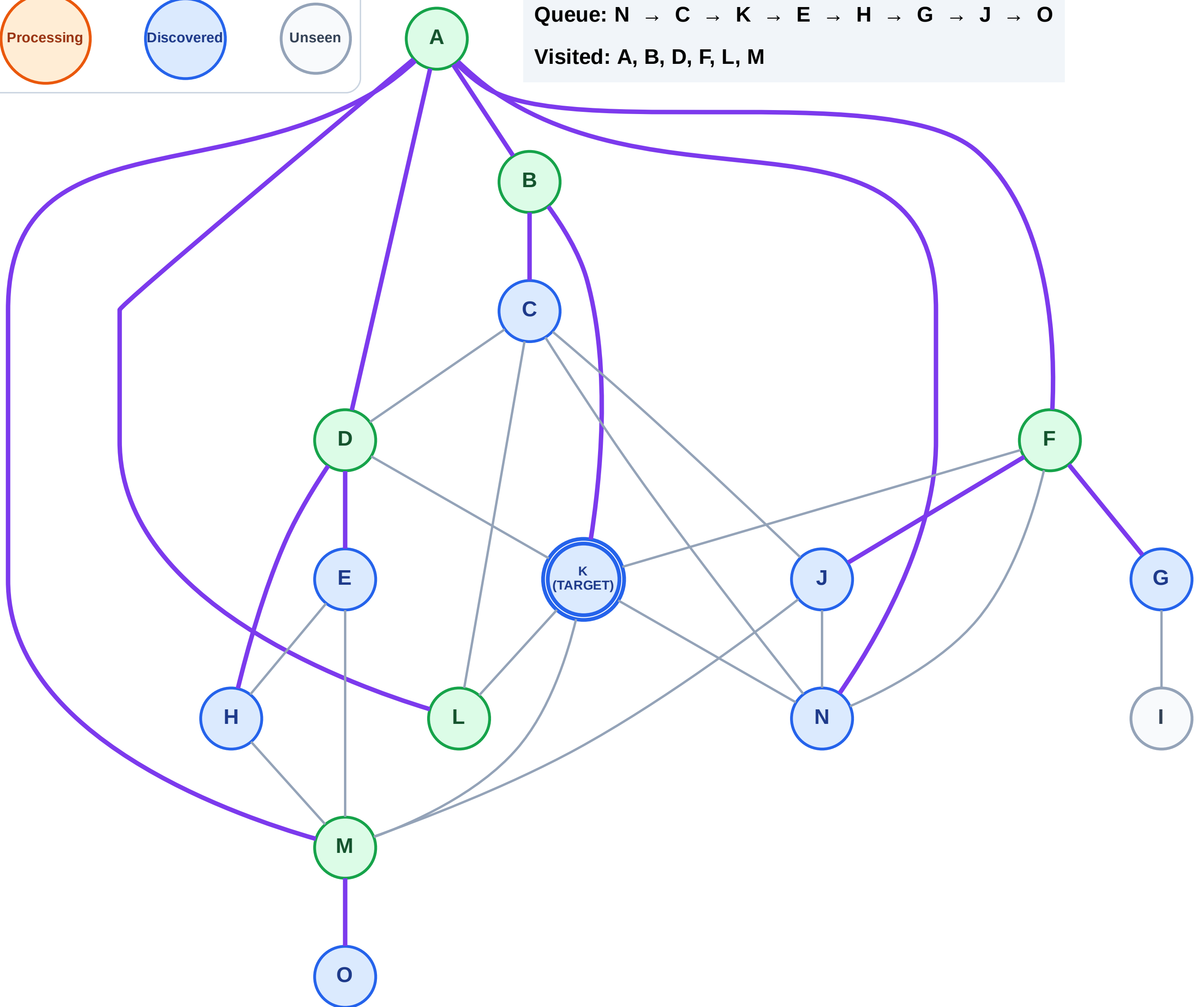
# BFS Traversal

Step 13: Discover O from M



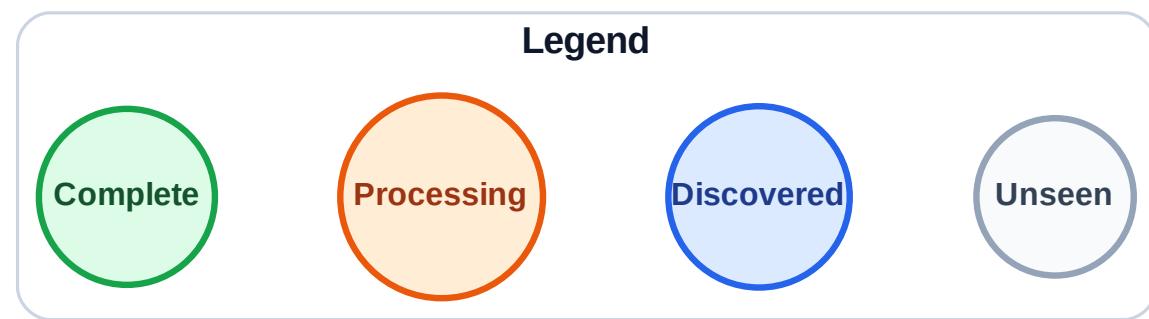
Queue: N → C → K → E → H → G → J → O

Visited: A, B, D, F, L, M



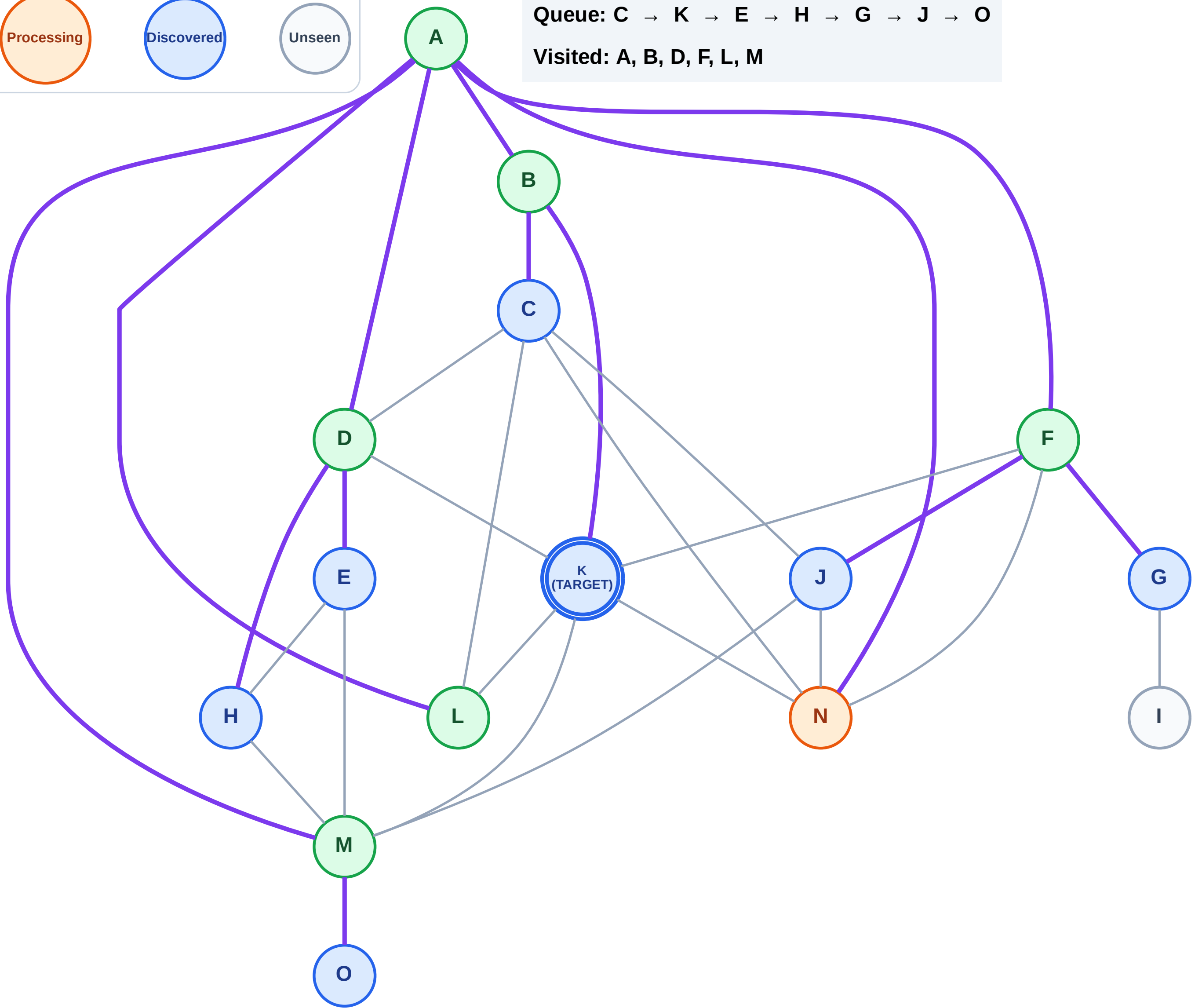
# BFS Traversal

Step 14: Process node N



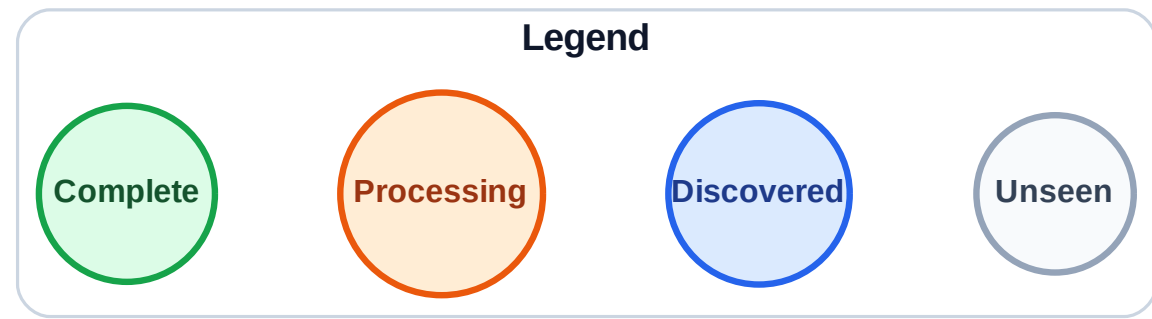
Queue: C → K → E → H → G → J → O

Visited: A, B, D, F, L, M



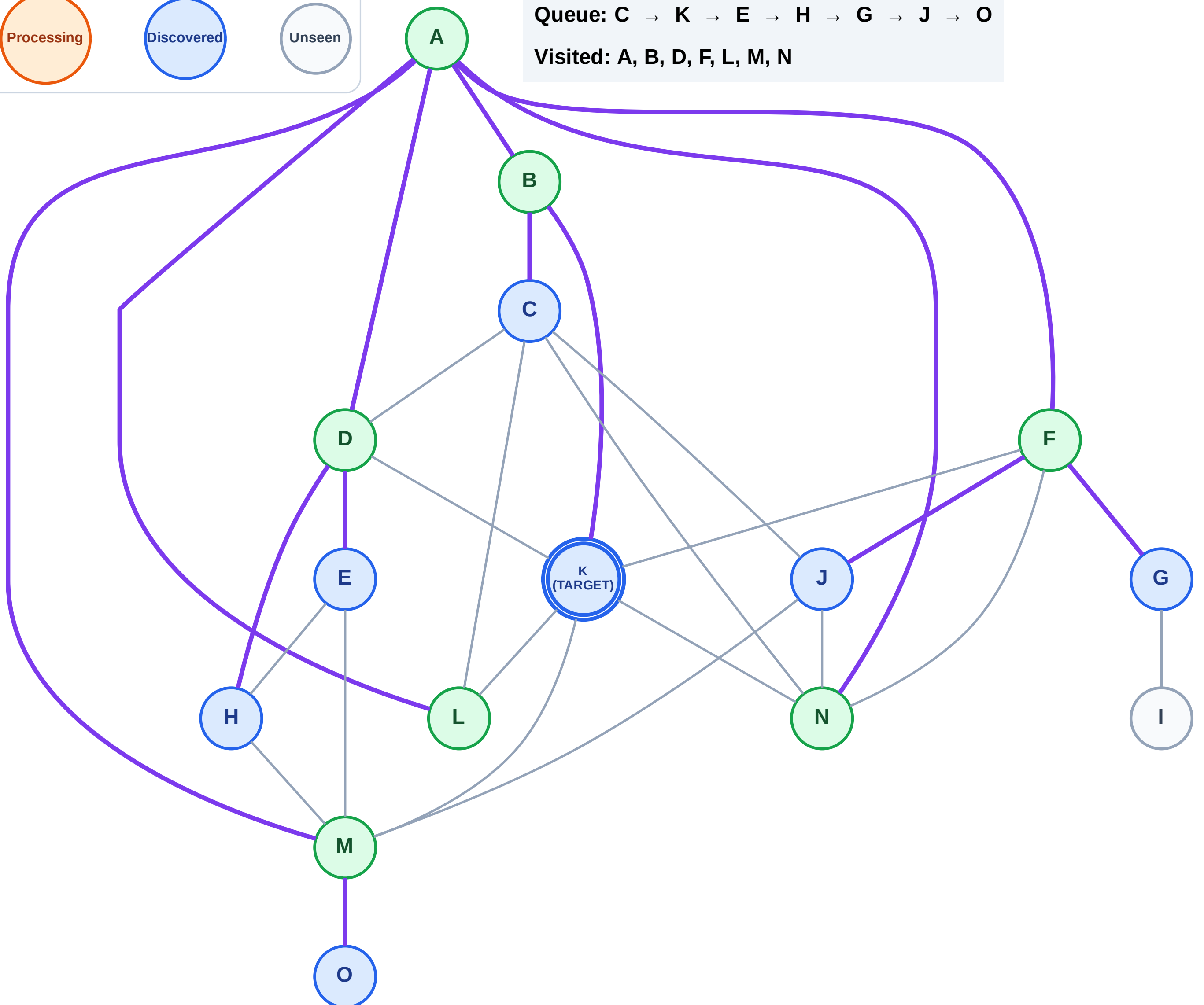
# BFS Traversal

Step 15: No new neighbors from N



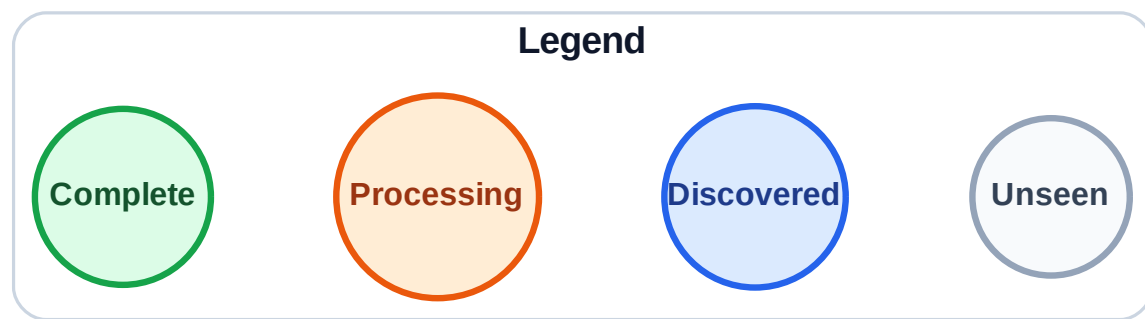
Queue: C → K → E → H → G → J → O

Visited: A, B, D, F, L, M, N



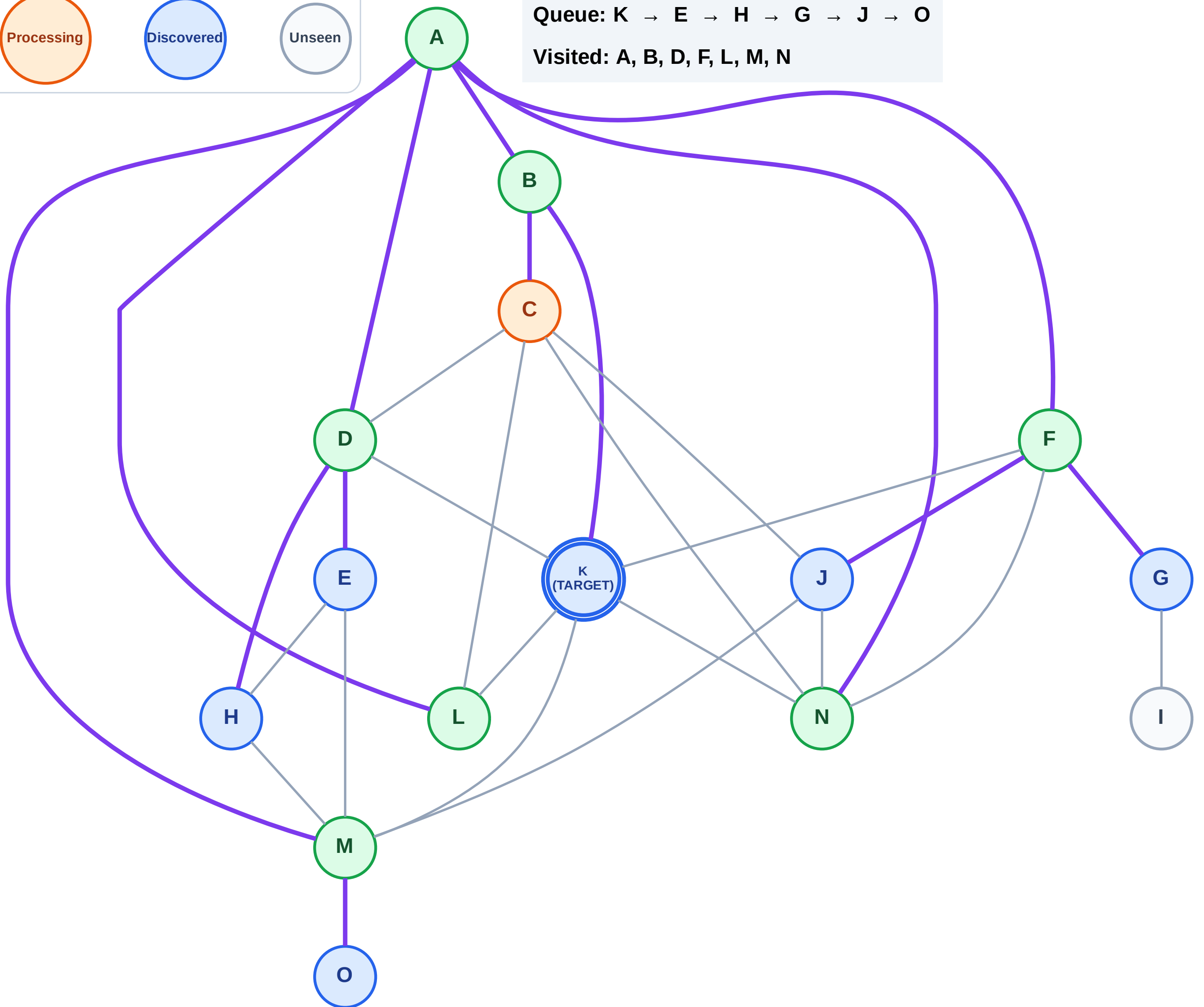
# BFS Traversal

Step 16: Process node C



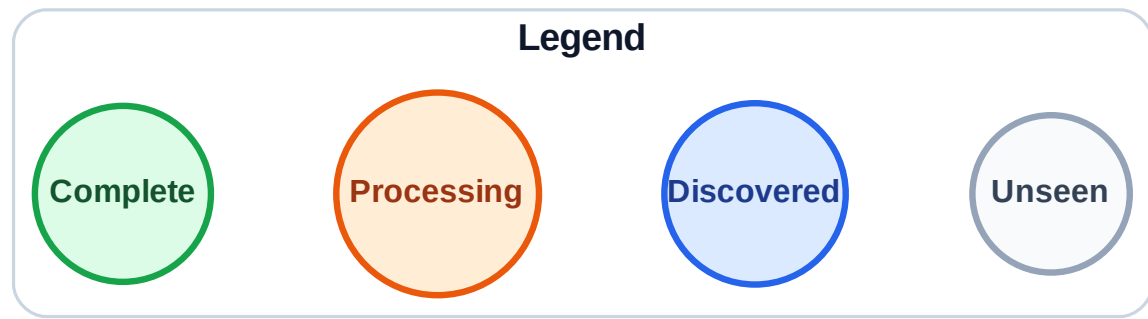
Queue: K → E → H → G → J → O

Visited: A, B, D, F, L, M, N



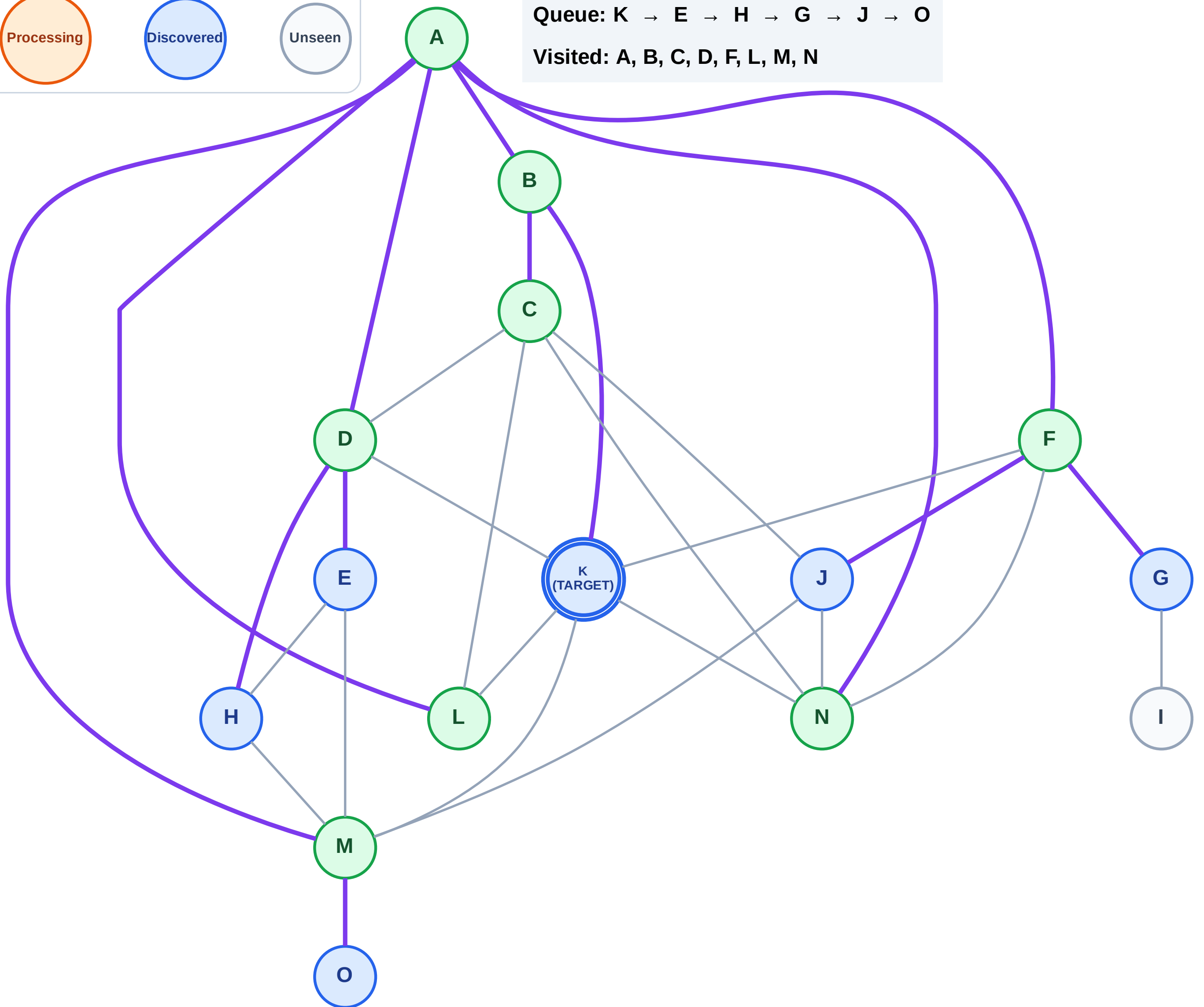
# BFS Traversal

Step 17: No new neighbors from C



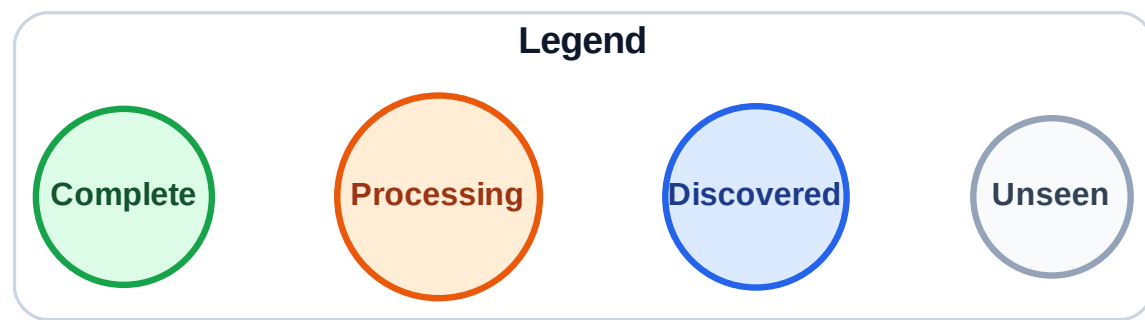
Queue: K → E → H → G → J → O

Visited: A, B, C, D, F, L, M, N

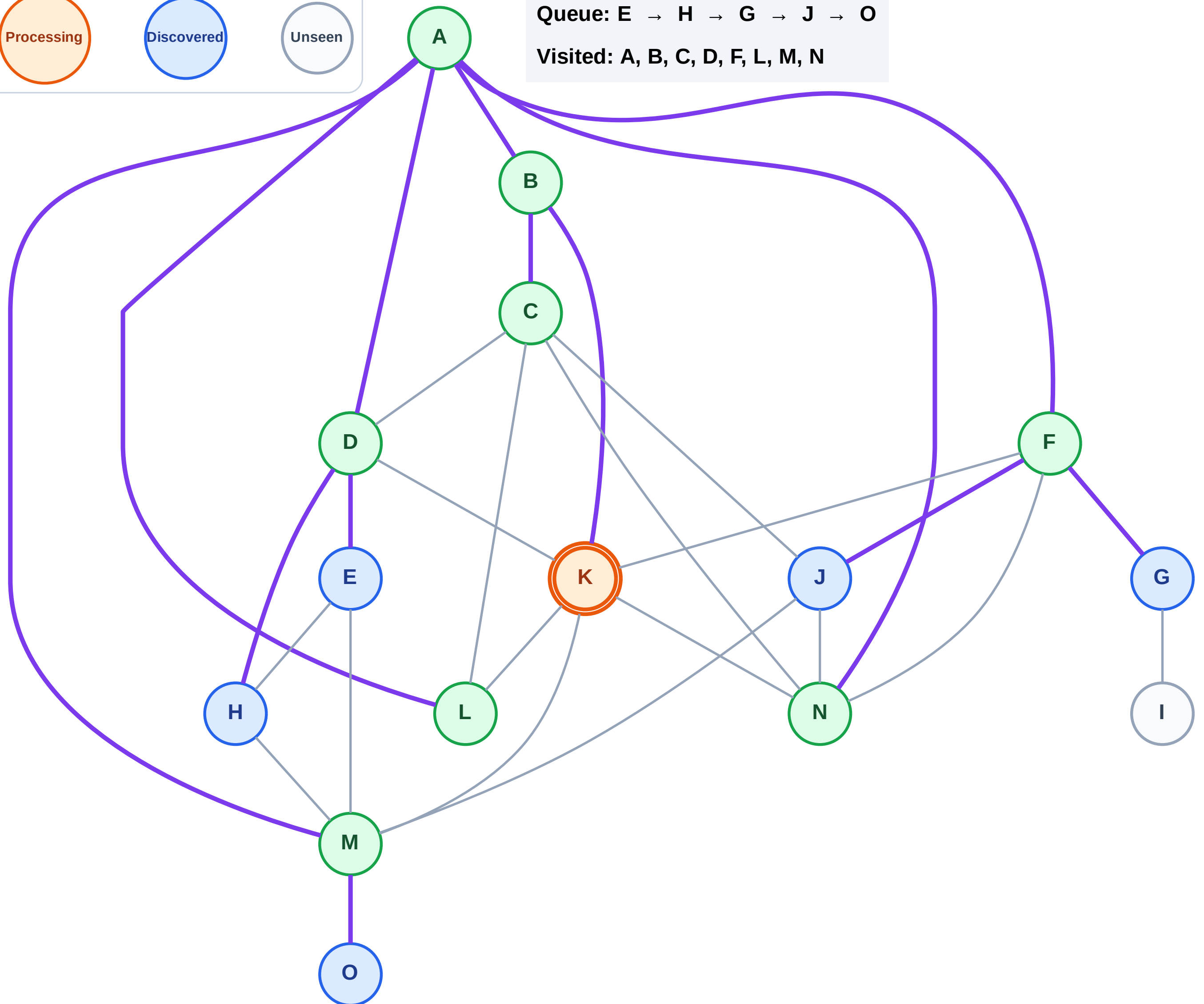


# BFS Traversal

Step 18: Process node K



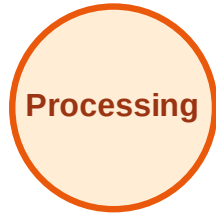
Queue: E → H → G → J → O  
Visited: A, B, C, D, F, L, M, N



# BFS Traversal

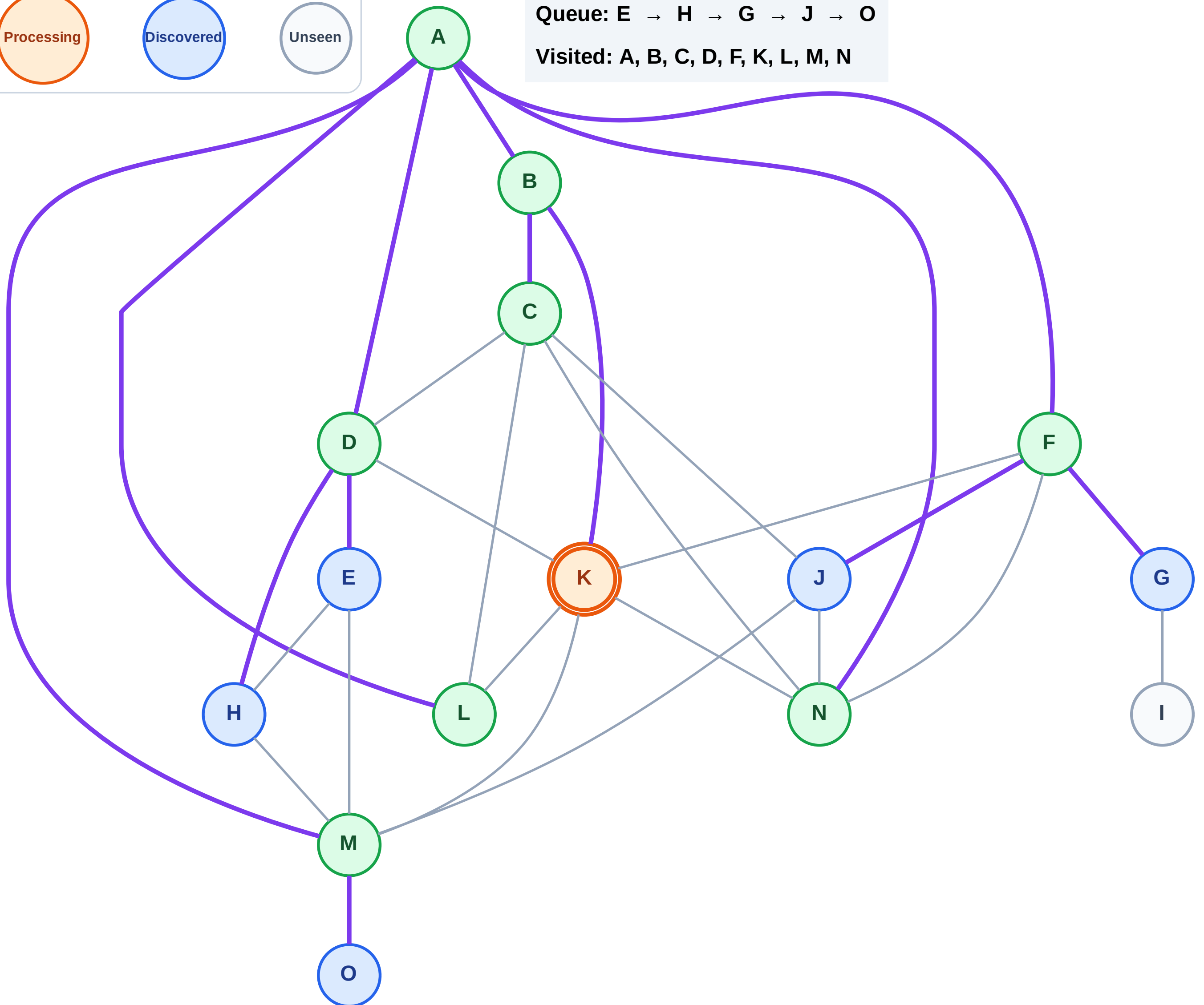
Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

## Legend



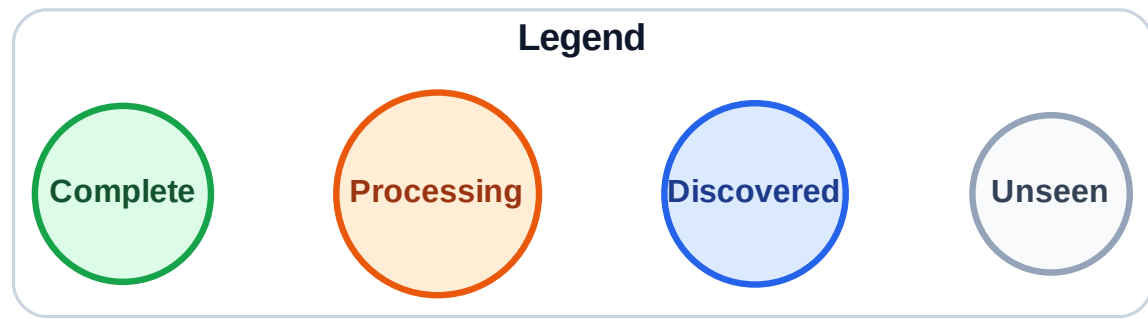
Queue: E → H → G → J → O

Visited: A, B, C, D, F, K, L, M, N



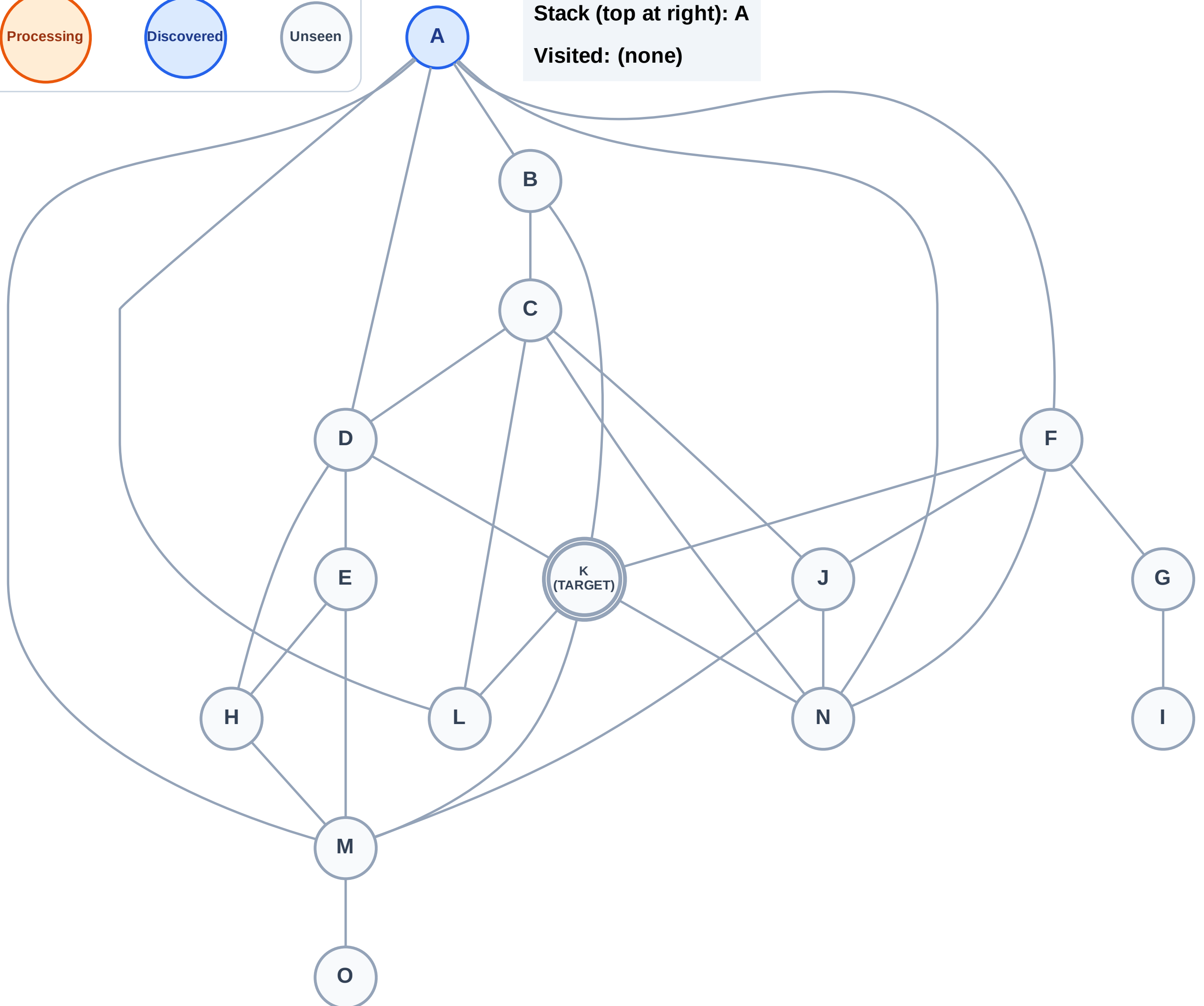
# DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

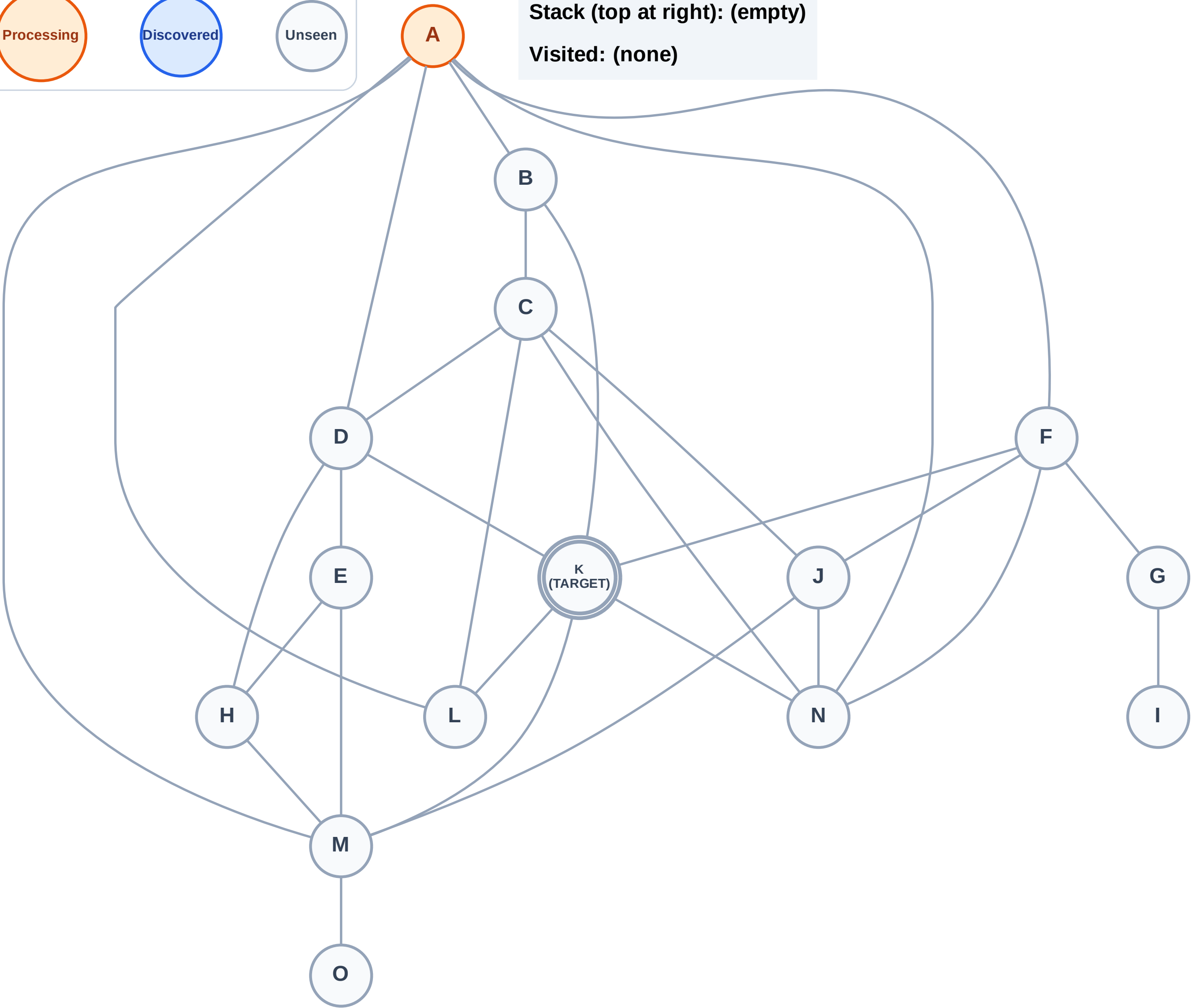
Complete

Processing

Discovered

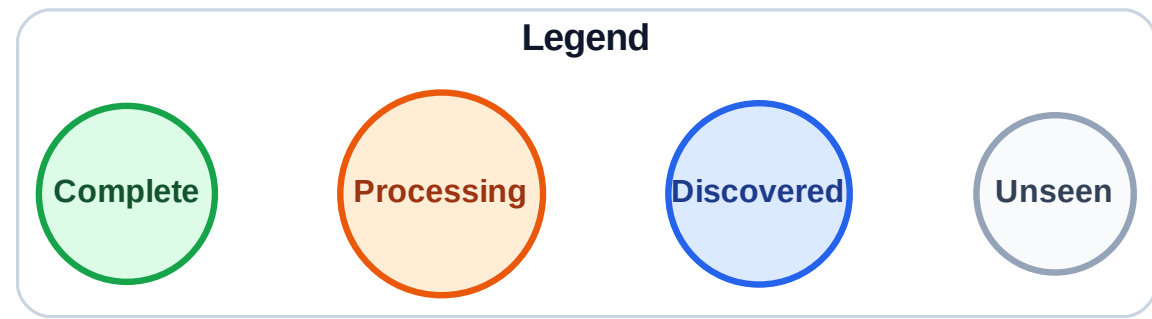
Unseen

Stack (top at right): (empty)  
Visited: (none)



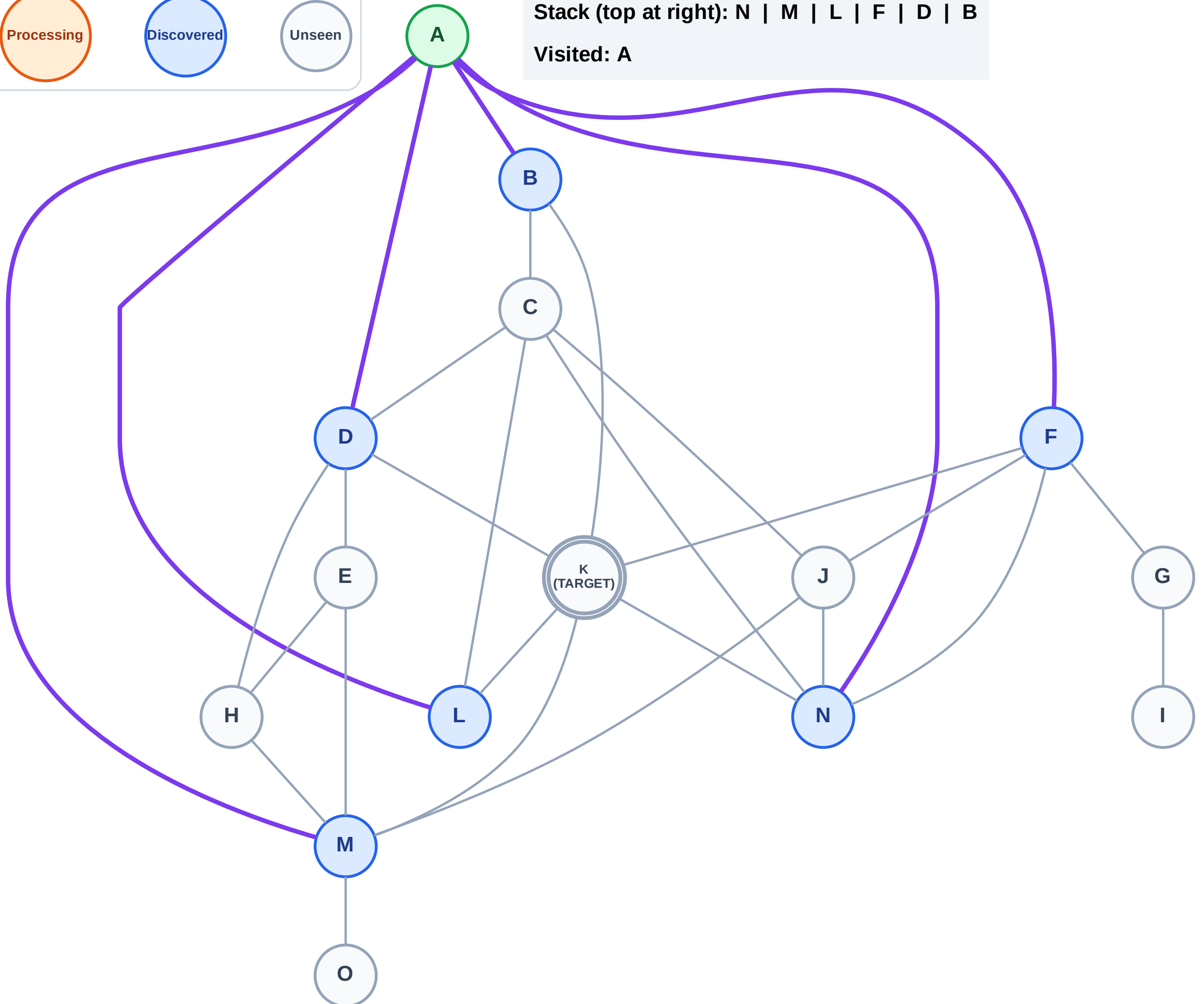
# DFS Traversal

Step 3: Discover N, M, L, F, D, B from A



Stack (top at right): N | M | L | F | D | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

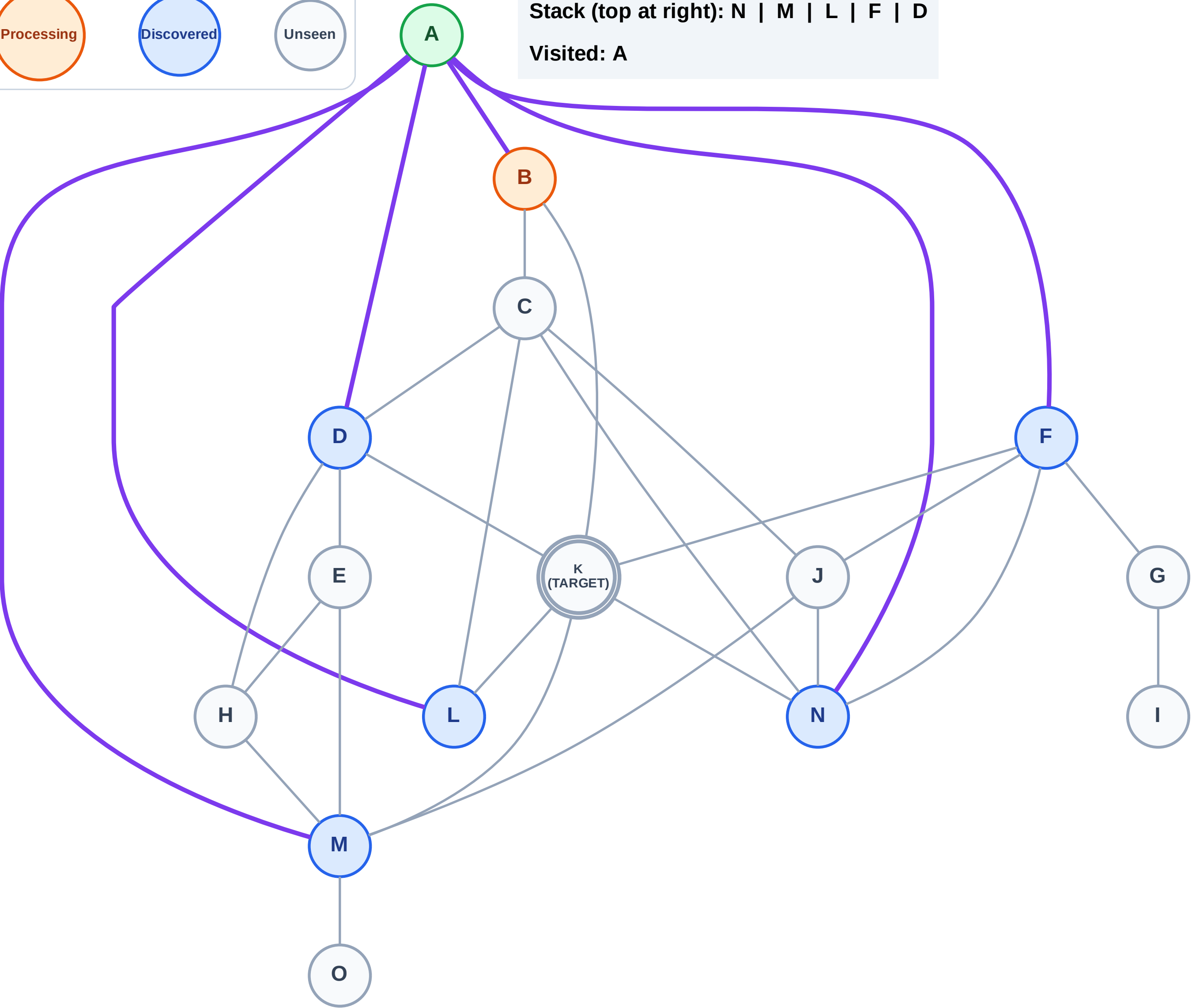
Processing

Discovered

Unseen

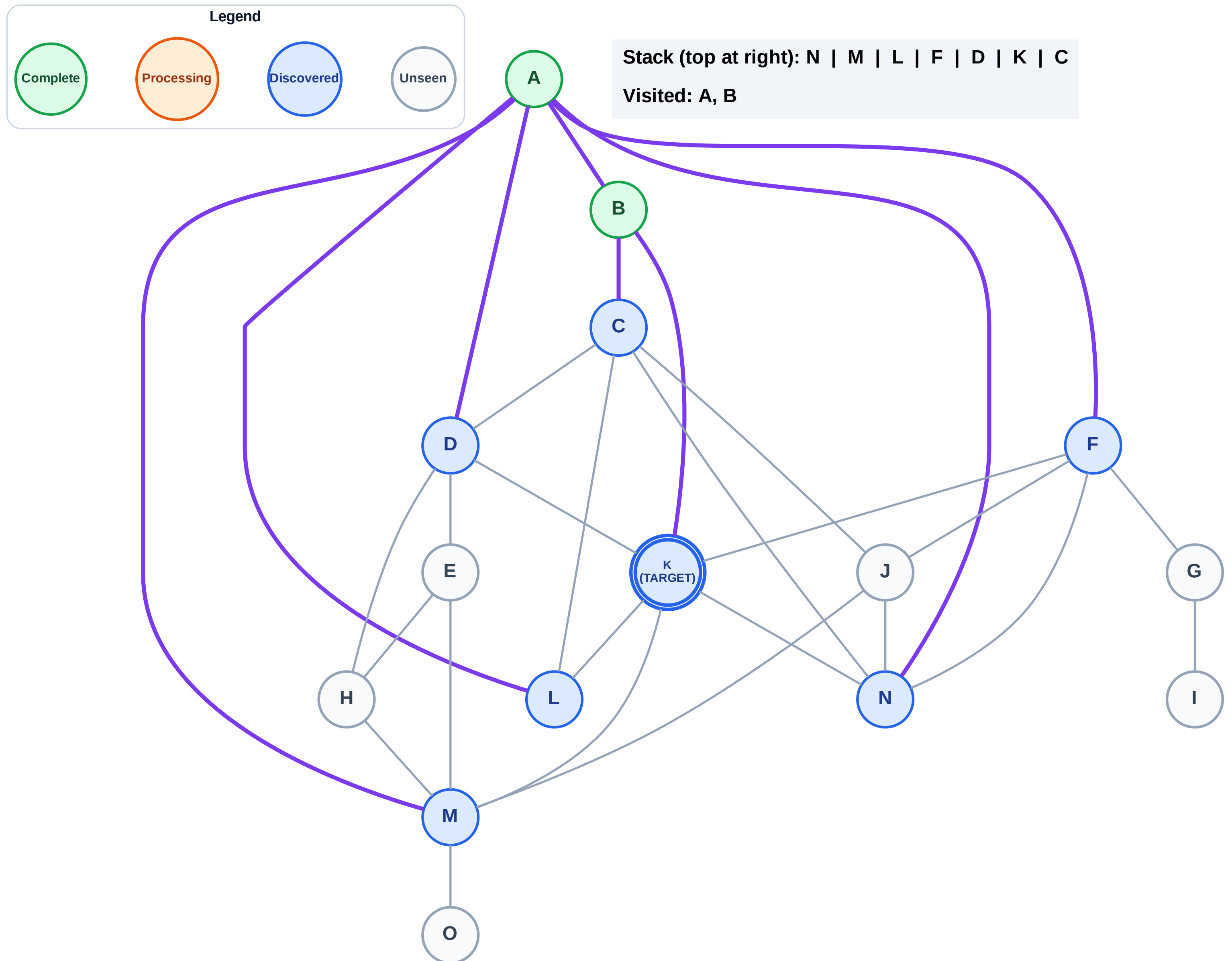
Stack (top at right): N | M | L | F | D

Visited: A



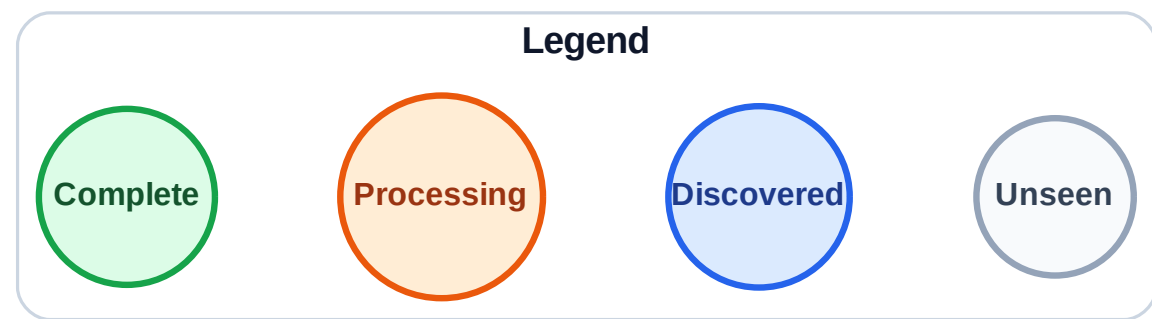
### Step 5: Discover K, C from B

### Step 5: Discover K, C from B



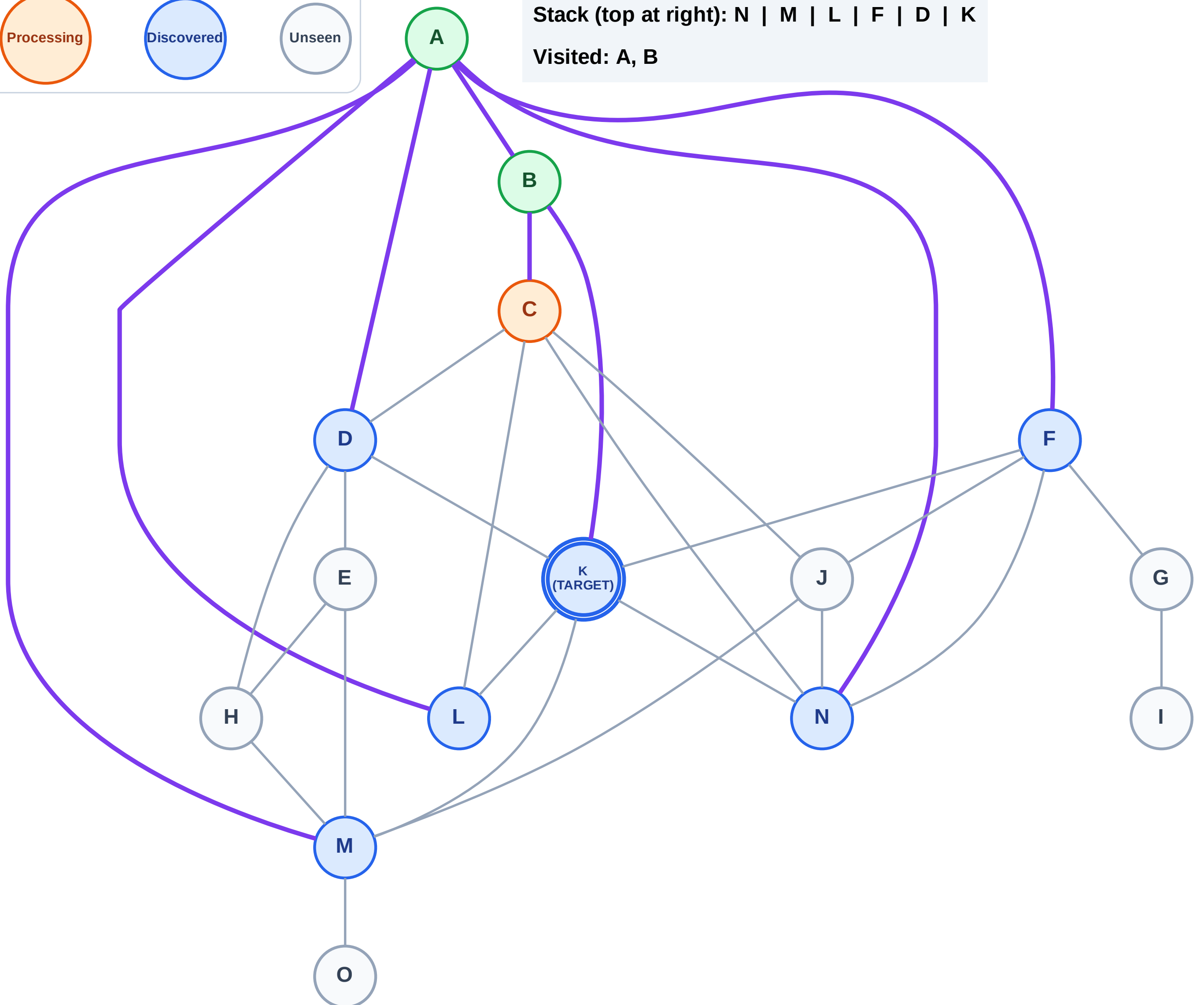
# DFS Traversal

Step 6: Process node C



Stack (top at right): N | M | L | F | D | K

Visited: A, B



# DFS Traversal

Step 7: Discover J from C

Legend

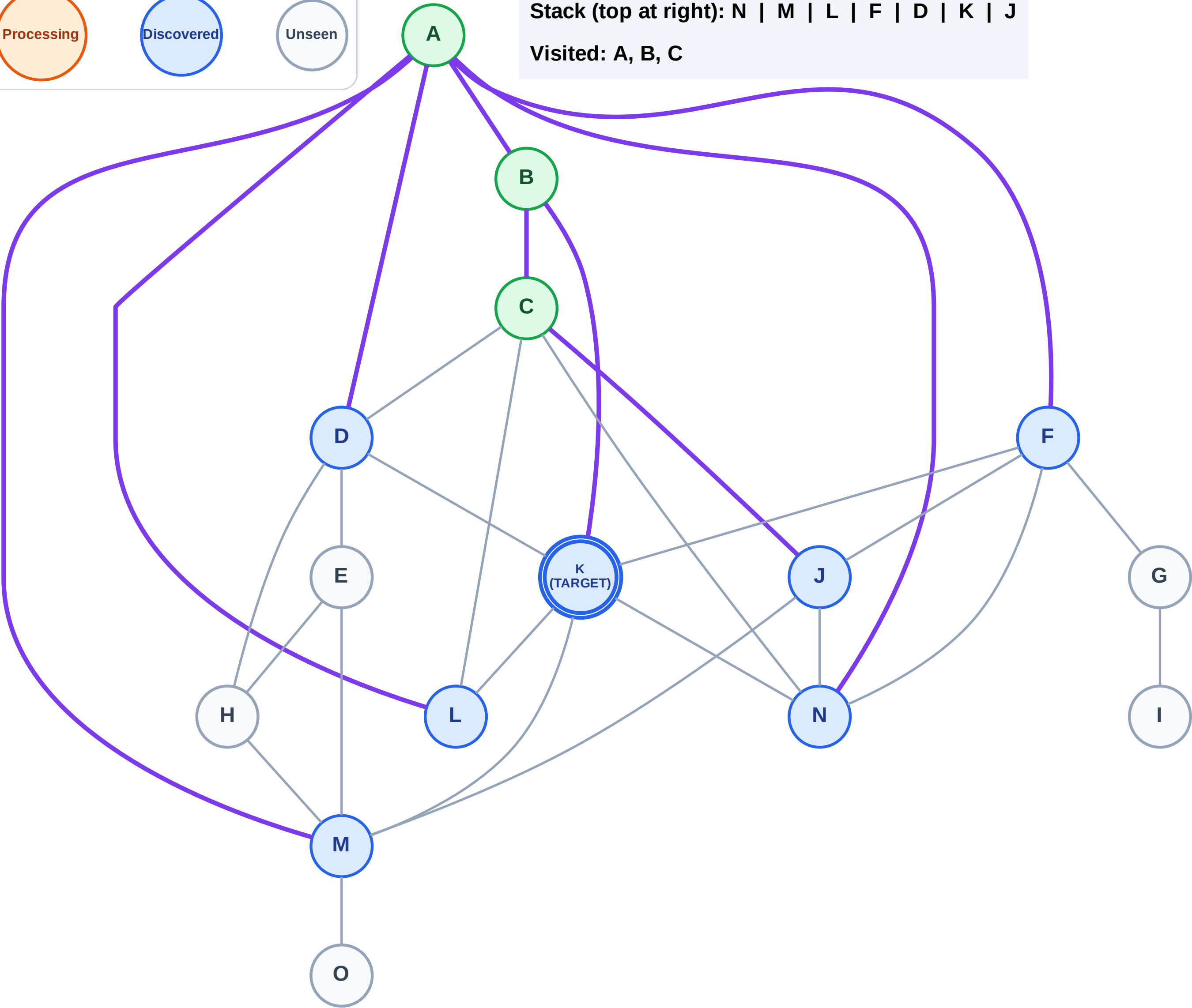
Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | L | F | D | K | J  
Visited: A, B, C



# DFS Traversal

Step 8: Process node J

Legend

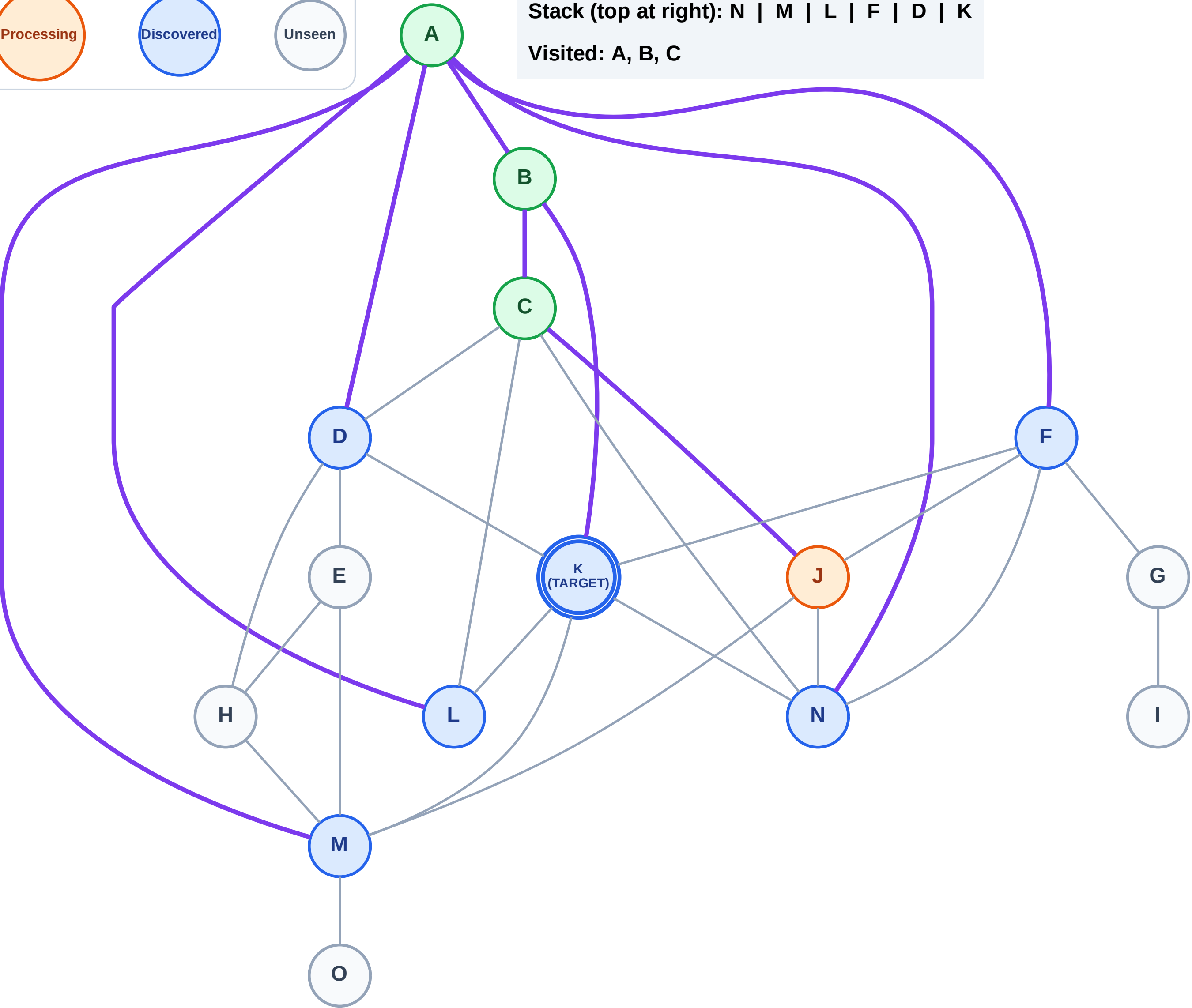
Complete

Processing

Discovered

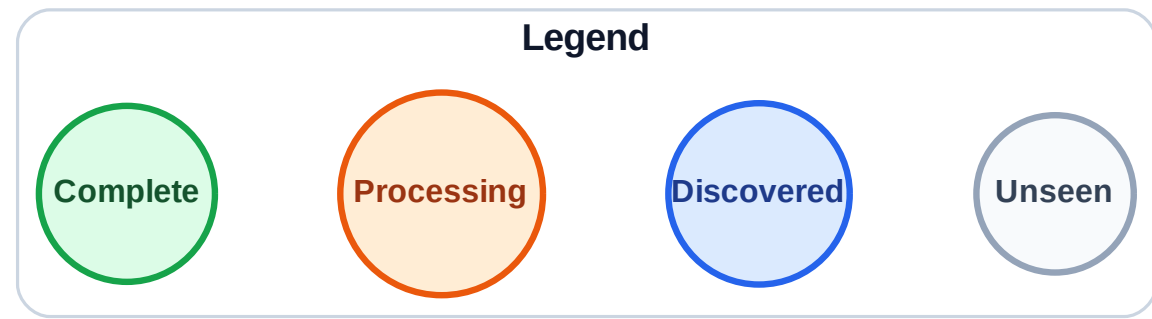
Unseen

Stack (top at right): N | M | L | F | D | K  
Visited: A, B, C



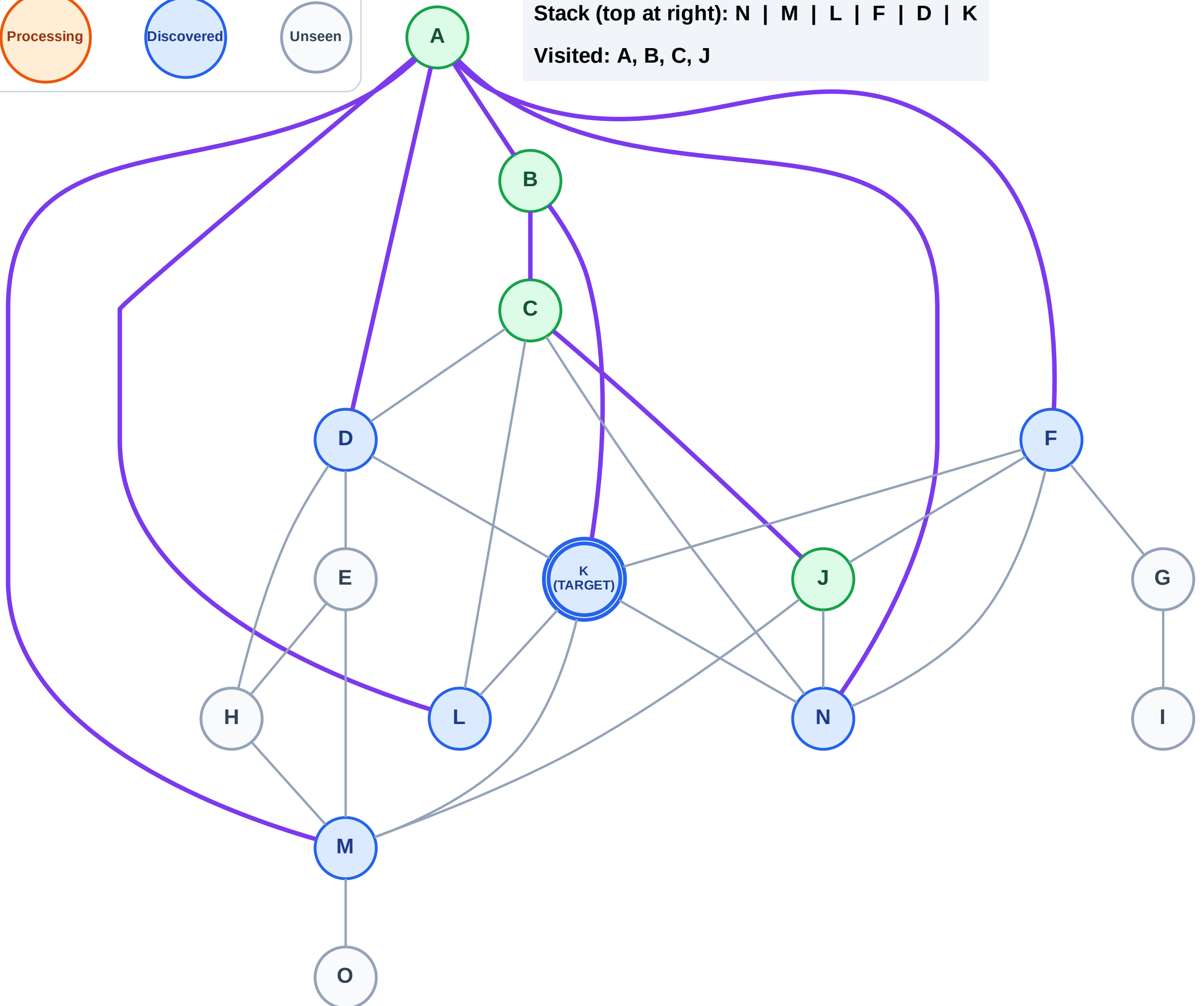
# DFS Traversal

Step 9: No new neighbors from J



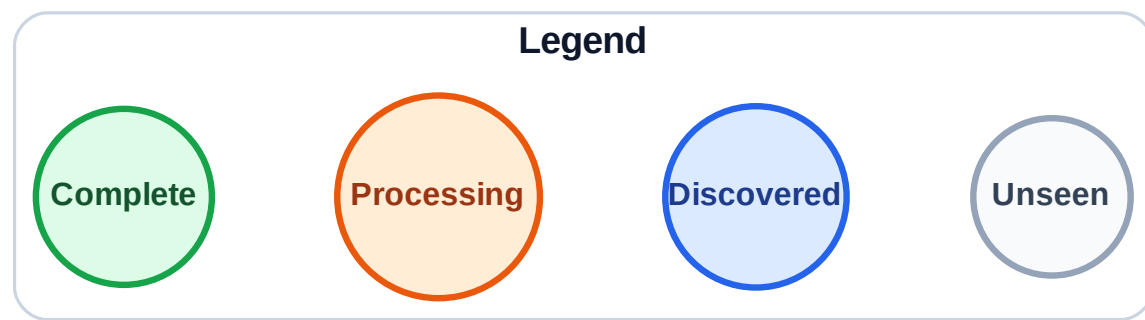
Stack (top at right): N | M | L | F | D | K

Visited: A, B, C, J

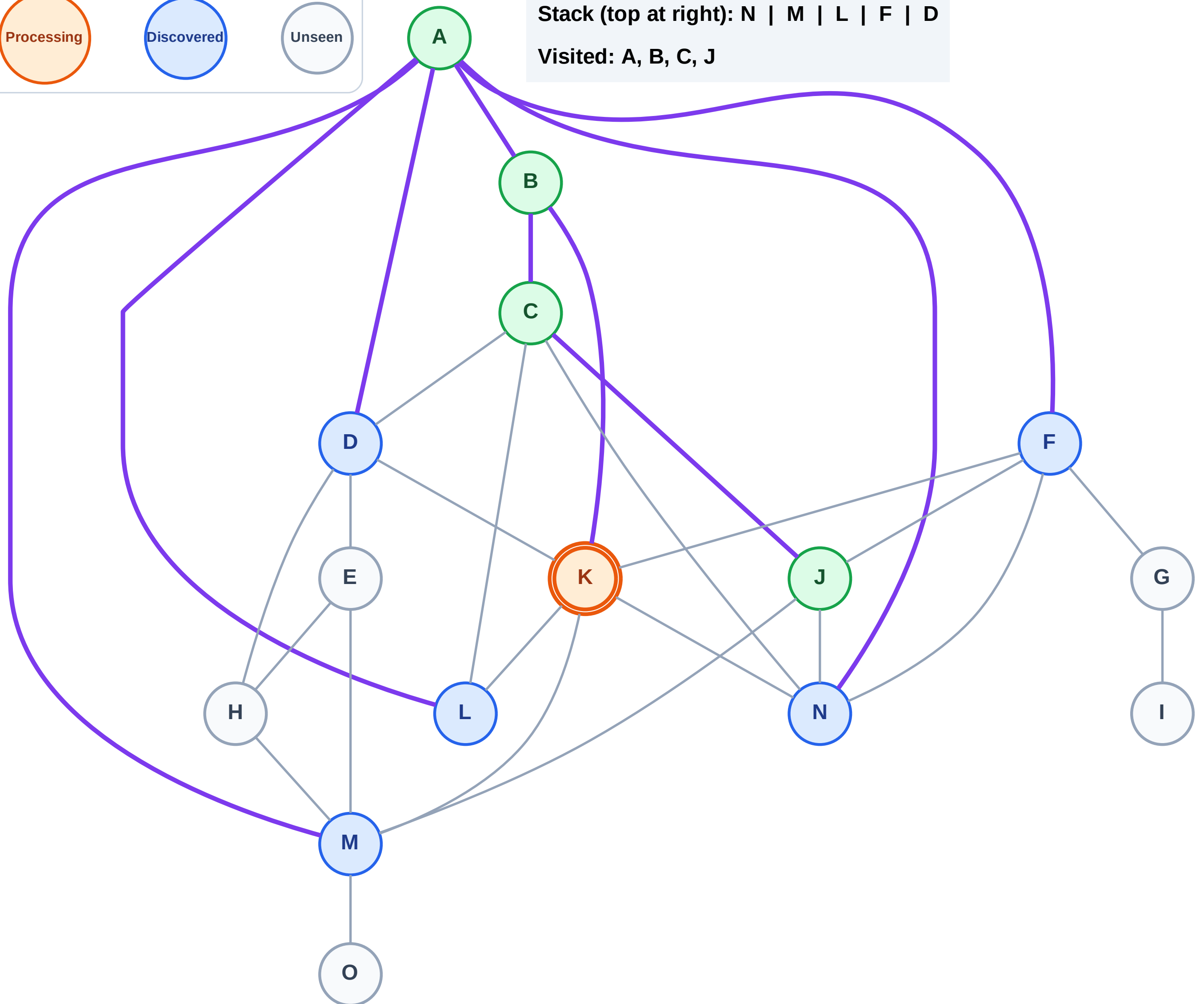


# DFS Traversal

Step 10: Process node K

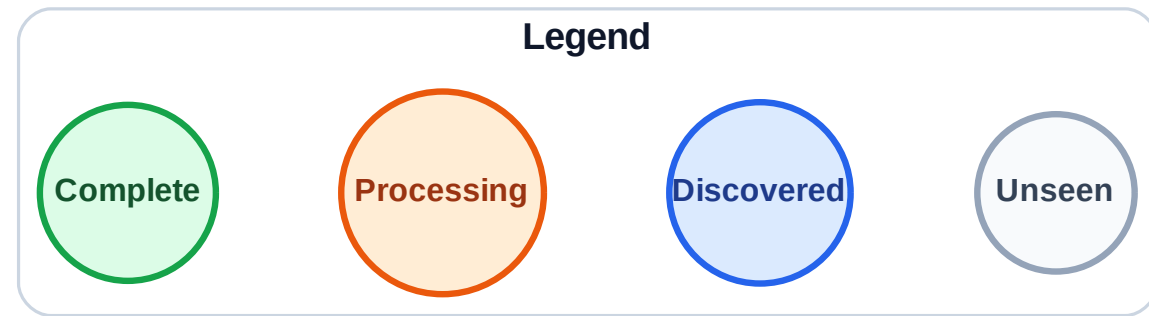


Stack (top at right): N | M | L | F | D  
Visited: A, B, C, J



# DFS Traversal

Step 11: TARGET NODE K REACHED! Path (2 hops): A → B → K



Stack (top at right): N | M | L | F | D

Visited: A, B, C, J, K

